Active-Rail Spacetime Metric Engineering System
Technical Disclosure and Source-Placement Architecture

Primary Operating Embodiment: Service Factor $V = 5$
Edge-Load Operating Embodiment: Service Factor $V = 10$

Document purpose. This document discloses a representative active-rail spacetime metric engineering architecture, a source-placement control structure, a source-family target architecture, and the diagnostic acceptance criteria used to characterize the design. The disclosure is organized in a patent-style technical format and emphasizes system components, operating modes, metric components, source ledgers, source-target roles, causal carrier diagnostics, and affirmative operating criteria for source-family realization.

1. ABSTRACT

An active-rail spacetime metric engineering system is disclosed. The system includes a prepared standing support substrate, a protected live packet corridor, a support-contained carrying-flow field, timed entry/catch/release choreography, source-family target accounting, endpoint receiver accounting, regulated endpoint source-medium closure, causal-carrier diagnostics, and chronology-governed service sequencing. The architecture separates baseline infrastructure support from service-induced transport demand and expresses the reduced service geometry in ADM variables for packet safety, source placement, radial-null margin, branch-band transport, exterior-null service-time rating, and reset restoration. A dimensionless service factor V specifies the carrying-flow operating load and supplies a common rating parameter for source ledgers and operational A-to-B timing ledgers. The resulting design combines packet-safe service choreography, explicit source-family assignment, finite-difference endpoint source closure, carrier-band finite-bundle accounting, rail-time service governance, and operational exterior-endpoint timing evaluation.

2. TECHNICAL FIELD

The disclosure relates to prescribed spacetime metric engineering, numerical relativity scaffolding, source-placement architectures, and controlled metric-service systems. More particularly, the disclosure relates to a rail-like spacetime metric in which prepared infrastructure bears broad geometric support while live service demand is routed through localized handoff windows and packet-adjacent control regions.

3. ENGINEERING OBJECTIVE

A practical spacetime metric engineering architecture benefits from explicit source placement, explicit service timing, and explicit causal-carrier classification. The disclosed system assigns standing radial support, throat/angular support, clock-lapse safety, carrying-flow load, entry service gating, catch/rematch correction, packet release, endpoint receiver action, support-shell timing, and reset operations to distinct service components. The engineering objective is to provide a reduced metric and source-placement process that:

1. separates prepared substrate support from live transport increments;
2. keeps the protected packet corridor causally safe throughout live service;
3. localizes small service-induced momentum demand into catch/rematch and support-shell handoff regions;
4. expresses the reduced metric in ADM variables for constraint-ledger diagnosis;

5. subtracts the standing substrate from live service fields to isolate transport demand;
6. characterizes 4D demanded-source responses for load-bearing component assignment;
7. supports prescribed reduced geometry, source-family target realization, endpoint source-medium closure diagnostics, causal-carrier auditing, and coarse 3+1 constraint accounting within one design grammar.

4. OVERVIEW OF THE ACTIVE-RAIL SYSTEM

The active rail operates as a coordinated metric service. A prepared standing substrate supplies the baseline throat, support shell, angular jacket, rail stretch, and clock-lapse structure. A live packet corridor lies inside the support shell. A carrying-flow field expresses the commanded service load through the prepared support infrastructure. The entry service gate assigns setup support to infrastructure accounting and begins live packet scoring inside the supported entry corridor. A locked timing layer coordinates support-side catch with packet rematch. A release-choreography layer then holds the packet briefly in a matched state, fades beta through a finite smooth profile, keeps lapse/carve support active through the handoff, and relaxes infrastructure support after the packet edge transfers the principal mismatch into the designated handoff windows. Packet-edge and support-shell control components are applied during catch/rematch and release to address localized handoff demand.

The invention-level structure is the componentized service architecture together with its source-target grammar and carrier classification. The service factor V is an operating-load parameter used to rate embodiments, select support/cushion/timing settings, compare source-demand scaling, and reconstruct A-to-B service-time advantage relative to an exterior null signal. The primary operating embodiment is $V = 5$. Lower-V cases, such as $V = 2$, provide diagnostic microscopes for mechanism and margin. The $V = 10$ case supplies an edge-load embodiment for the same architecture. The representative V5 family is packet-safe under smoothing and grid refinement. Demanded-source burden is structured: live angular/current peaks are assigned to packet-in-support entry rows, while non-live radial-null and radial-pressure burdens are carried by support-plant infrastructure. The compact handoff layer reduces broad semilocal radial-null and angular-pressure concentration by using endpoint-smooth entry, catch, and edge windows, while pressure-aware entry shaping preserves a clear control channel for live radial pressure. The source-family target interprets the principal non-live radial body as a constant-flux radial string-cloud scaffold and realizes the endpoint/junction layer as a finite non-live endpoint source package: a bounded reset-decompression cap, a finite support-edge closure component, a small endpoint current regulator, and a regulated anisotropic heat/current medium whose internal angular response closes the beta075 endpoint source at finite-difference field-closure level. The carrier classification records local GZ-like branch behavior as a controlled service-carrier feature and evaluates service protection through reachability, radial escape, scheduled probes, dense branch-band bundle transport, wake-tail restoration, exterior-null service-time rating, and rail-time chronology governance.

5. REPRESENTATIVE DESIGN ARCHITECTURE

The representative design has five coupled layers. The metric-service layer is the prepared ADM program $(\alpha, \beta, \gamma_{ll}, \gamma_{\Omega\Omega})$ with $V = 5$ as the primary operating rating. The packet-handoff layer is the matched-hold release choreography with a smooth beta fade, blended trailing-edge beta-rematch sleeve, finite beta-collar widening for branch-band bundle preservation, two-feature packet-local radial support, compact entry/catch/edge handoff support, and entry service gate near $\sigma = -1.40$. The source-family layer is

$$\mathcal{S}_{\text{target}} = S_0 + J + R + C_{\text{live}} + G + D_H, \tag{1}$$

where S_0 is the constant-flux radial string-cloud backbone, J is the coupled endpoint/junction layer, R is small core-body residual leakage, C_{live} is live handoff trim, G is angular-capacity support, and D_H is non-live current relaxation.

The endpoint receiver/source layer uses beta-release memory to drive a localized negative-l angular receiver in the non-live support-edge band. The beta075 $p003\_mid$ receiver is the representative effective receiver: it preserves the live packet gate, improves endpoint selected-null transfer, and retains positive affine-reparameterized radial-null benchmark margin in dense source-sector accounting. The associated beta075 endpoint source is frozen as a bounded reset-cap body plus a finite support-edge closure component. Its radial heat/current block is represented by a regulated anisotropic heat/current medium with a small non-live current regulator, and its non-live angular-pressure watch is carried as an internal angular response of the same medium. The beta100 edge-load condition uses the same receiver grammar with service-specific timing; a mild post-release retiming anchor supplies positive selected, current, and angular transfer accounting on the beta100 stress grid.

The causal-carrier layer evaluates the shell/throat carrying-flow feature through a dedicated prescribed-metric carrier audit. The local GZ-like branch signature is measured in the live and shell/throat overlap bands. Service-stage and carry-stage entry-to-live-packet reachability screens record zero hits in the sampled ledgers. Refined beta075 live packet, main carrier, support plant, packet geometry, and live branch-band seeds reach exterior radial boundaries on both radial null branches, including minus-branch traces through the branch-sensitive region. Dense finite-bundle branch-band audits are included with the single-ray and scheduled-probe gates: the selected widened trailing-edge beta-rematch collar keeps dense branch-band bundles radially escaping, recovered from both-shrinking intervals, and above the caustic-like collapse thresholds used in the prescribed-metric audit. The service-time advantage ledger compares the reconstructed restored packet arrival event with an exterior null reference between the same endpoints and rates the representative schedule as operationally favorable. The chronology-governance layer treats each permitted service edge as an advance in a shared rail time with readiness margin. Reset-domain wake-tail monitors remain exterior to live service.

6. DEFINITIONS

Term	Meaning in this disclosure
Active rail	A spacetime metric service in which prepared infrastructure supports the transport geometry and a live packet is carried through a protected corridor.
Service factor V	A dimensionless carried-flow service setting. It sets the scheduled plateau level of the support-contained carrying-flow field, the packet-frame comparison schedule used for causal-margin accounting, and the direct service-factor proxy used in A-to-B service-time rating.
Service-time advantage ledger	An operational exterior-null comparison ledger. It fixes exterior endpoints A and B, defines packet departure and restored arrival events, reconstructs service time through entry, carried shift, catch/release, and restoration, and compares that service time with an ordinary exterior null reference time.
Prepared standing plant accounting	A timing convention in which already prepared infrastructure is accounted as standing plant, while request-triggered setup is included in the service-time clock.
Rail time T_{rail}	A service-governor chronology variable assigned to an active rail or connected rail network. Admissible service edges, ordinary causal edges, reset intervals, and return/cross-link edges advance this variable with configured safety margin.

Chronology-governed service sequencing	A service-arming rule in which entry, carried shift, catch/release, endpoint readiness, reset, and network routing are admitted according to monotone T_{rail} advance. It frames closed-timelike-curve exposure as a future safety-engineering task for service scheduling and network governance.
Primary operating embodiment	The representative service configuration evaluated at $V = 5$.
Edge-load operating embodiment	The representative service configuration evaluated at $V = 10$ for the same architecture.
Diagnostic low-service check	A lower service-factor case, such as $V = 2$, used to inspect mechanism, causal margin, and whether a timing law benefits from explicit V-awareness.
Entry service gate	A service-accounting boundary that assigns pre-entry setup support to infrastructure accounting and begins live packet scoring inside the supported entry corridor.
Standing substrate	Prepared support geometry including throat/core support, radial support shell, angular jacket, and baseline lapse structure.
Live packet	Passenger-facing corridor whose causal and source-exposure properties are monitored during live service.
Support shell	Wider infrastructure region enclosing the live packet and carrying broad support demand. In the support-shell control overlays, the active support-shell band is an annular infrastructure band at the support edge.
Carrying-flow field	The service-facing name for the radial shift component β. It carries the scheduled service load through prepared support infrastructure.
Clock-lapse field	The service-facing name for the lapse component α. It supplies causal margin and can be used as a timed partner to carrying-flow control.
Rail-stretch field	The service-facing name for the radial spatial metric component $G = \gamma_{ll}$. It sets radial metric capacity and affects how carrying-flow gradients appear in the source ledger.
Throat-capacity field	The service-facing name for the angular metric component $Q = \gamma_{\Omega\Omega}$. It includes throat and angular-jacket capacity.
Locked timing	Timing rule in which support-side catch begins with a controlled lead relative to packet rematch while packet-frame safety is constrained.
Release choreography	Coordinated matched-hold, beta fade, lapse support, carve support, and packet-edge rematch sequence used to remove carrying-flow support while controlling derivative cost in the live packet.
Matched hold	A release-choreography phase in which the packet is kept locally matched for a short interval while the trailing edge and support infrastructure settle.
Trailing-edge beta-rematch sleeve	A packet-adjacent beta correction layer, strongest near the inner/trailing packet edge and weak in the packet center, used to soften local $(v_{\mathrm{coord}} + \beta)$ mismatch while preserving a quiet packet center.

Finite minus-branch carrier-focus collar	The localized branch-band handoff layer in which the minus radial null family is most sensitive to beta-rematch tuning. In the representative design it is treated as a controlled carrier-rematch collar outside the passenger-facing transport corridor.
Beta-collar rematch lead	A widened trailing-edge beta-rematch collar that preserves packet timelikeness and dense branch-band bundle width in the prescribed-metric carrier audit. The conservative representative lead is the width-6.0, temporal-1.5 trailing-edge rematch setting, with a width-8.0, temporal-2.0 setting retained as a wider optical-relief bracket.
Blended beta floor	A smooth union between a weak filled packet-center beta floor and a shaped trailing-edge rematch sleeve. The representative V5 operating embodiment uses this blended floor with a widened sleeve.
Two-feature radial support profile	A packet-local γ_{ll} support profile using a mild filled core stretch, a narrow annular catch ring, and a weaker annular skirt. It reduces radial-null and angular-pressure burden while preserving packet safety.
Catch/rematch absorber	Localized packet-edge source or control component assigned to residual null/momentum handoff demand.
Radial support edge	Transition layer controlling radial pressure demand and support-shell coupling.
Lapse cushion	Clock-lapse-sector compensator that preserves causal margin consumed by stronger support-edge shaping.
Support-shell overlay	A localized metric-side support-shell component applied through a smooth timing and radial window $W_{\rm sh}(\sigma, l)$. V5 source-ledger studies identify raised-cosine annular support-shell bearing as a clean comparator and reusable control knob; the release-choreography family retains support-shell/lapse infrastructure while focusing selection on packet-edge handoff smoothing.
Compact handoff layer	An endpoint-smooth entry/catch/edge support layer that separates entry pressure containment from broad catch/rematch and packet-edge smoothing.
Pressure-aware entry profile	A compact entry support function that carries live radial-pressure containment while preserving a separate handoff sleeve for angular/current and radial-null derivative relief.
Semilocal null-accumulation diagnostic	A radial-null source diagnostic that averages T_{kk} over finite service/radial windows and compares the result with a selected quantum-inequality-style floor.
Affine-reparameterized radial-null diagnostic	A finite-window radial-null screen using an explicitly reparameterized affine window coordinate. It computes radial-null non-affinity, integrates the associated affine weight, and records radial geodesic residual monitors.
4D demanded-source ledger	Finite-difference evaluation of $T^{\rm dem}_{\mu\nu} = G_{\mu\nu}[g]/(8\pi)$ for the prescribed metric. It diagnoses radial null, radial pressure, angular pressure, current, and packet-comoving density demands.
Strong 4D source redistribution	A source-ledger result in which a metric component measurably reduces or reroutes expensive demanded-source channels while preserving packet safety, bounded point peaks, and numerical robustness.

Standing substrate subtraction	Diagnostic subtraction of a prepared beta-off or ready-state substrate from live service fields to isolate service-induced ADM demand.
Source-family target	A source-side realization target derived from the demanded-source ledger. It assigns demanded stress roles to admissible source sectors and, for the beta075 repaired lead, is paired with a finite endpoint source-medium closure package.
Constant-flux radial string-cloud backbone	The non-live radial-tension scaffold assigned to the dominant infrastructure radial support. It has positive energy density, radial tension close to $p_l = -\rho$, small current, small transverse pressure, and nearly constant areal flux proxy $\mu = \Phi/\gamma_{\Omega\Omega}$.
Endpoint/junction layer	A coupled non-live source target at support-edge and reset/decompression endpoints. It carries residual radial trim, angular capacity, and current relaxation after subtracting the constant-flux radial backbone.
Beta-memory receiver	A support-edge endpoint receiver whose amplitude is tied to accumulated beta-release transfer and expressed as an active endpoint receiver channel. The representative effective receiver uses a negative-l localized angular channel in the endpoint/junction grammar.
Frozen endpoint-J source	The beta075 repaired-lead endpoint source package: a bounded reset-decompression cap body paired with a finite support-edge closure component. It is non-live, finite-width, and stable under the baseline and dense source ledgers used for the current disclosure.
Endpoint current regulator	A small non-live regulator in the $\rho+p_l$ radial pair that makes the endpoint radial heat/current block boost-diagonalizable without adding current, angular, collar, packet, or extra conservation-closure components.
Regulated anisotropic heat/current medium	The current beta075 endpoint physical-source candidate. It carries the regulated radial heat/current block, keeps the heat-flux frame subluminal with positive margin, and carries the remaining non-live angular-pressure watch as an internal anisotropic response.
Constrained medium field closure	A finite-difference validation in which the frozen endpoint source, endpoint current regulator, and internal angular response close the beta075 endpoint medium without live leakage or added design components.
Causal carrier diagnostic	A radial ADM diagnostic using $v_{\pm} = -\beta \pm \alpha/\sqrt{\gamma_{ll}}$, branch-margin maps, measured cone/branch signatures, reachability masks, escape traces, and scheduled probe evolution to characterize controlled one-way carrier behavior and live-service radial escape.
Measured GZ-like cone signature	A measured shell/throat carrier-band feature in which a radial null branch crosses zero in the sampled ADM chart while $g_{\sigma\sigma}$ changes sign in the active support region. It is classified by reachability, radial escape, and scheduled probe gates.
CHL-adjacent branch-band optics	A comparison class for null-geodesic access and cone-like optical behavior in shift-driven warp geometries, following Clark, Hiscock, and Larson. In this disclosure it identifies a finite-bundle optical comparison regime for branch-band null bundles while preserving the active rail as a throat-loaded service-carrier architecture.

Dense finite-bundle carrier audit	A prescribed-metric null-bundle diagnostic that fans strong recovered branch-band seeds into dense fixed-time radial bundles and measures escape, recovery from both-shrinking intervals, bundle-width ratios, ray ordering, and caustic-like collapse flags.
Scheduled ADM probe audit	A prescribed-metric probe evolution through $\alpha(\sigma, l)$, $\beta(\sigma, l)$, $\gamma_{ll}(\sigma, l)$, and $\gamma_{\Omega\Omega}(\sigma, l)$ for packet centerline, packet edge, radial null, off-axis null, congruence, and reset-tail monitors.
Wake-tail restoration monitor	A reset-domain diagnostic for packet-geometry minus-branch tails after release. It records exterior clearing, live re-entry, and future-domain closure separately from live-service carrier acceptance.

7. OPERATING REGIME AND ROLE OF V

The service factor V is the operating rating assigned to the carrying-flow sector of the reduced active-rail metric. It is a natural focus parameter because one scalar sets the plateau level of the scheduled carrying-flow service before the architecture distributes that service through support-side catch, packet rematch, radial support, and release/fade envelopes. Holding the geometric support parameters fixed while varying V therefore tests how a selected rail architecture responds to a higher or lower commanded carrying-flow load.

The scheduled support-side carrying-flow factor and the packet-frame comparison factor are obtained by blending V with an exit or residual service factor v_{exit} through the locked timing schedules:

$$U_\beta(\sigma) = v_{\text{exit}} + (V - v_{\text{exit}})\, C_\beta(\sigma), \tag{2}$$

$$U_{\text{pkt}}(\sigma) = v_{\text{exit}} + (V - v_{\text{exit}})\, C_{\text{pkt}}(\sigma), \tag{3}$$

where C_β is the support-side catch schedule and C_{pkt} is the packet rematch schedule. In the locked-timing setting, C_β begins with a controlled support-side lead while C_{pkt} is closely coupled by packet-norm safety. Thus V selects the service level, while the choreography determines where and when that service level is expressed.

The carrying-flow factor enters the representative metric through the carrying-flow field β. A compact form of the support-contained carrying-flow ansatz is

$$\beta(l, \sigma) = -\frac{U_\beta(\sigma)\, E(\sigma)\, W(l)^{p_\beta}\, S(l, \sigma)}{Q(l, \sigma)}, \tag{4}$$

where E is the release/fade envelope, W is the support-shell radial window, S is the packet-adjacent service envelope, and $Q = \gamma_{\Omega\Omega}$ is the throat-capacity field. The packet-frame comparison uses

$$u^l_{\text{pkt}}(\sigma, l) = \frac{U_{\text{pkt}}(\sigma)}{Q(l, \sigma)}, \qquad \mathcal{N}_{\text{pkt}} = -\alpha^2 + G\left(u^l_{\text{pkt}} + \beta\right)^2, \tag{5}$$

with packet causal safety requiring $\mathcal{N}_{\text{pkt}} < 0$ on the live packet corridor. In this sense, V drives the metric service along the path

$$V \to U_\beta, U_{\text{pkt}} \to \beta, \mathcal{N}_{\text{pkt}} \to K_{ij}, \rho_{\text{ADM}}, j_l, \tag{6}$$

so a V-sweep probes the same architecture's carried-flow load, packet causal margin, and ADM source-placement response.

In representative diagnostics, $V = 5$ is the primary operating embodiment. It provides a clean platform for ADM translation, standing-substrate subtraction, catch/rematch source-law characterization, release choreography, compact handoff shaping, source-family accounting, endpoint receiver accounting, and causal-carrier acceptance. Lower service factors, including $V = 2$, inspect mechanism and margin. The $V = 10$ embodiment exercises the same architecture at an edge-load rating. The representative V5 operating embodiment is characterized on V5 packet safety and source placement. The beta075 receiver setting is the promoted effective receiver regime, while beta100 supplies the release-timing stress boundary and retiming anchor for stronger beta-release conditions.

7.1 Operational Service-Time Rating

The service factor also supplies an operational exterior-null comparison. The service-time ledger fixes exterior endpoints A and B, defines the packet departure event at A and the restored arrival event at B, and compares the reconstructed service time with an ordinary exterior null reference. In units with $c = 1$, the exterior null reference time is

$$T_{\text{ext,null}} = D_{AB}. \tag{7}$$

The prepared-service time is

$$T_{\text{service}} = \sigma_{\text{arrive}} - \sigma_{\text{depart}}, \tag{8}$$

and the two transport proxies used in the current ledger are

$$D_{\text{sched}} = \int_{\sigma_{\text{depart}}}^{\sigma_{\text{arrive}}} U_{\text{pkt}}(\sigma)\, \mathrm{d}\sigma, \tag{9}$$

$$D_{\text{coord}} = \int_{\sigma_{\text{depart}}}^{\sigma_{\text{arrive}}} \frac{U_{\text{pkt}}(\sigma)}{B(\sigma)}\, \mathrm{d}\sigma, \tag{10}$$

where $B(\sigma)$ is the centerline throat/carrying-flow capacity factor recorded in the source ledger. The advantage ratios are $D_{\text{sched}}/T_{\text{service}}$ and $D_{\text{coord}}/T_{\text{service}}$. The strict source-grid centerline control uses the modeled tube relation $l = \sigma$ and remains unity by construction.

On the $s = 15$ service-time ledger, the representative schedule has prepared-service time approximately 3.045, schedule-factor distance approximately 7.822491, and packet-coordinate distance approximately 3.755436. The corresponding operational advantage ratios are approximately 2.568962 and 1.233312, respectively. With request-triggered setup included in the service clock, the same endpoint comparison remains favorable with ratios approximately 2.487278 and 1.194097. Thus the prescribed schedule has a positive exterior-null service-time rating while packet safety, dense finite-bundle transport, source-family realization, and global causal-structure scope remain assigned to their own diagnostic gates.

7.2 Chronology-Governed Service Sequencing

Operational service-time advantage brings the standard superluminal chronology issue into the service-safety envelope. Everett and Roman showed the network form of the issue clearly for Krasnikov tubes: a single directed tube has one-way superluminal behavior, while a pair of tubes can be arranged as a time-machine system [3]. Shoshany and Snodgrass give a modern two-warp construction in which linked warp-drive geometries form a closed timelike geodesic [4]. In the active-rail design, the known closed-timelike-curve problem is therefore treated as a safety-engineering problem for service arming, endpoint readiness, reset timing, and network routing.

The engineering handle is monotonicity in a rail-governed time. Let T_{rail} be a service chronology variable associated with a prepared rail or with a connected rail network. A permitted service edge $A \to B$ satisfies

$$T_{\text{rail}}(B_{\text{arrive}}) \geq T_{\text{rail}}(A_{\text{depart}}) + \Delta_{\text{safety}}, \tag{11}$$

where Δ_{safety} covers reset recovery, catch/release recovery, endpoint readiness, clock uncertainty, relative-motion margin, and multi-link routing margin. Return edges, cross-links, and ordinary exterior causal edges are evaluated in the same T_{rail} accounting. A multi-leg service cycle is admitted after the accumulated rail-time advance satisfies

$$\sum_{e\in C} \Delta T_{\mathrm{rail}}(e) > \Delta_{\mathrm{cycle}}, \tag{12}$$

for the configured cycle margin Δ_{cycle}.

The staged active-rail sequence supplies the control variables needed for this governor. Entry, carrier plateau, catch/release, endpoint receiver action, reset/decompression, and wake-tail restoration are distinct service states. Transport is armed during accepted rail-time service intervals and held in readiness or reset state during chronology-sensitive endpoint windows. A connected service network extends the same rule across stations and rails: all participating rails share one chronology governor or a certified translation into one global ordering. Network-scale synchronization, boosted-frame endpoint policies, and margin allocation are reserved as a future safety-engineering task.

8. REPRESENTATIVE REDUCED METRIC

A representative reduced metric is expressed as

$$\mathrm{d}s^2 = -\alpha(l,\sigma)^2\mathrm{d}\sigma^2 + G(l,\sigma)\left(\mathrm{d}l + \beta(l,\sigma)\mathrm{d}\sigma\right)^2 + Q(l,\sigma)\mathrm{d}\Omega^2, \tag{13}$$

where l is the rail coordinate and σ is the service scheduling coordinate. The shift sector is intentionally retained in the completed-square form. This presentation displays the active-rail geometry as a Morris–Thorne-like intrinsic throat/support geometry, represented by G and Q, with a shift-like carrying-flow sector β applied along the rail coordinate. The four metric-side service fields are:

$$\alpha(l,\sigma) \quad \text{clock-lapse field,} \tag{14}$$

$$\beta(l,\sigma) \quad \text{carrying-flow field,} \tag{15}$$

$$G(l,\sigma) = \gamma_{ll} \quad \text{rail-stretch field,} \tag{16}$$

$$Q(l,\sigma) = \gamma_{\Omega\Omega} \quad \text{throat-capacity field.} \tag{17}$$

This notation reserves distinct symbols for metric components and support-shell overlay amplitudes.

8.1 General Component Form

The representative metric family can be described as a shaped baseline plus localized support-shell partners:

$$\beta(l,\sigma) = \beta_0(l,\sigma) + a_\beta W_{\mathrm{sh}}(l,\sigma), \tag{18}$$

$$\alpha(l,\sigma) = \alpha_0(l,\sigma)\exp\left(\kappa_\alpha W_{\mathrm{sh}}(l,\sigma)\right), \tag{19}$$

$$G(l,\sigma) = G_0(l,\sigma)\exp\left(\kappa_G W_{\mathrm{sh}}(l,\sigma)\right), \tag{20}$$

$$Q(l,\sigma) = Q_0(l,\sigma)\exp\left(\kappa_Q W_{\mathrm{sh}}(l,\sigma)\right). \tag{21}$$

The baseline fields $\alpha_0, \beta_0, G_0, Q_0$ include the prepared throat, angular jacket, radial support-edge softening, lapse cushion, entry service gate, shaped catch/rematch timing, service release, packet-edge beta rematch, packet-local radial support shaping, compact entry/catch/handoff support, and packet-frame comparison structure. The localized partners are support-shell metric overlays. The continuous partner family includes carrying-flow (a_β), clock-lapse (κ_α), rail-stretch (κ_G), and throat-capacity (κ_Q). Support-shell source-ledger accounting assigns carrying-flow plus clock-lapse as the default partner pair, rail-stretch as a peak-shaping comparator, and same-window throat-capacity as neutral in the default setting. The representative V5 operating embodiment uses release choreography, a blended trailing-edge packet beta-rematch sleeve, a two-feature packet-local radial support profile, and compact pressure-aware handoff support to preserve live packet safety and control demanded-source placement through the catch/release handoff.

The support-shell window is an annular, timed, packet-excluding window:

$$W_{\rm sh}(l,\sigma) = R_{\rm sh}(|l|)\, T_{\rm sh}(\sigma;\lambda_c,\tau_c)\, P_{\rm excl}(l,\sigma), \tag{22}$$

where $R_{\rm sh}$ selects support-edge infrastructure, $T_{\rm sh}$ places the control action relative to catch/rematch, and $P_{\rm excl}$ suppresses direct live-packet overlap. In the representative implementation,

$$0.65\, R_{\rm th} \le |l| \le 1.20\, R_{\rm th} \tag{23}$$

is the nominal support-shell band, λ_c is the support-shell catch lead, and τ_c is the temporal width of the support-shell timing envelope.

The preferred V5 support-shell source-ledger comparator uses a compact raised-cosine annulus centered in the nominal support-shell band:

$$R_{\rm sh}(r) = \frac{1}{2}\left[1+\cos\!\left(\pi\frac{|r-r_c|}{h_r}\right)\right], \qquad |r-r_c| \le h_r, \tag{24}$$

$$R_{\rm sh}(r) = 0, \qquad |r-r_c| > h_r, \tag{25}$$

where $r = |l|$, r_c is the midpoint of the active support-shell band, and the preferred half-width is approximately $h_r = 0.48125$ for $R_{\rm th} = 1.75$ with the above multipliers. Gaussian and smooth-box annular windows are retained as comparator shapes, and the raised-cosine annulus gives the cleanest source-ledger behavior in matched-strength V5 accounting.

The preferred timing envelope is Gaussian in the service coordinate and normalized against the active catch/rematch edge:

$$T_{\rm sh}(\sigma;\lambda_c,\tau_c) = \frac{\exp[-(\sigma-\sigma_c)^2/(2\tau_c^2)]}{\max\left(\exp[-(\sigma_*-\sigma_c)^2/(2\tau_c^2)],\epsilon_T\right)}, \qquad \sigma_c = \sigma_{\rm catch} - \lambda_c, \tag{26}$$

where σ_* is the closest point in the active catch/rematch band to σ_c. Minimum-jerk and raised-cosine temporal pulses are useful comparators. The representative timing band is $\tau_c = 0.30$–0.35 with $\lambda_c = 1.45$–1.55 for the V5 support-shell source-ledger family.

For matched-strength shape comparison, the support-shell carrying-flow overlay is parameterized by the peak shell perturbation

$$\Delta\beta_{\rm sh,max} = \max_{l,\sigma} |a_\beta W_{\rm sh}(l,\sigma)|\,. \tag{27}$$

This scalar makes radial and temporal shapes comparable at equal effective peak support-shell strength. The preferred V5 operating band is $\Delta\beta_{\rm sh,max} \approx 0.20$–$0.25$, with $\Delta\beta_{\rm sh,max} \approx 0.30$–$0.35$ used as an edge-stress range for point-peak and shell-throat concentration checks.

8.2 Release Choreography, Compact Handoff, and Packet-Edge Rematch

The representative V5 operating embodiment adds a release-choreography layer to the support-shell infrastructure. The release law is a coordinated handoff in which the packet is first caught/rematched, briefly held in a matched state, released through a finite smooth beta fade, and protected by lapse/carve support until packet-edge mismatch has been transferred into the designated handoff windows.

A representative release envelope is written as

$$W_{\rm rel}(\sigma) = \mathcal{S}_5\!\left(\frac{\sigma-\sigma_{\rm hold}}{\Delta\sigma_\beta}\right), \qquad \mathcal{S}_5(x) = 10x^3 - 15x^4 + 6x^5, \quad 0 \le x \le 1, \tag{28}$$

with constant matched hold before the finite fade and saturated values outside the fade interval. Higher-smoothness profiles remain comparator choices, but the present preferred release setting uses a minimum-jerk profile because its endpoint derivatives vanish and because it keeps the release law interpretable.

The compact handoff layer expresses the same release principle spatially. It uses separate endpoint-smooth windows for entry containment, catch/rematch bearing, and trailing-edge handoff smoothing:

$$W_{\mathrm{ch}}(l,\sigma) = g_e W_e(l,\sigma) + g_c W_c(l,\sigma) + g_h W_h(l,\sigma), \tag{29}$$

where W_e supplies broad live-entry pressure containment, W_c supplies catch/rematch bearing, and W_h supplies outer packet-edge derivative smoothing. A representative high-resolution V5 compact handoff uses a seventh-order smoothstep transition, an entry containment strength near 0.75, catch strength near 0.15, edge strength near 0.16, a broad edge width multiplier near 7.2, and an entry service gate beginning near $\sigma = -1.40$. This structure gives explicit control over radial pressure through the entry window while assigning angular/current and radial-null derivative relief to the catch and edge windows.

The entry service gate is a source-accounting boundary that represents setup physics in the full infrastructure ledger and begins live packet scoring once the packet has entered the supported corridor. In the representative high-resolution V5 ledger on the extended 101×151 domain, the entry-gated compact handoff has zero positive live packet-norm samples and a maximum live packet norm of approximately -1.2424.

The packet-edge beta rematch acts on the local mismatch

$$v_{\mathrm{coord}} + \beta_{\mathrm{pre}}, \tag{30}$$

where β_{pre} denotes the carrying-flow field before packet-local beta rematch. A compact notation for the packet-local correction is

$$\delta\beta_{\mathrm{pkt}}(l,\sigma) = -g_\beta\, W_{\beta,\mathrm{pkt}}(l,\sigma)\, \left(v_{\mathrm{coord}} + \beta_{\mathrm{pre}}(l,\sigma)\right). \tag{31}$$

The preferred $W_{\beta,\mathrm{pkt}}$ is a trailing-edge sleeve: weak in the packet center, strongest near the inner/behind side of the packet edge, and temporally active through the live catch/rematch corridor. The shaped sleeve is combined with a weak filled-core floor using a smooth blended union,

$$W_{\beta,\mathrm{pkt}} = W_{\mathrm{sleeve}} + W_{\mathrm{floor}}\left(1 - W_{\mathrm{sleeve}}\right), \tag{32}$$

which gives a single continuous packet-edge correction profile. The high-resolution V5 operating embodiment uses a beta-rematch gain of about 1.8, center floor 0.60, inner/outer sleeve radii approximately $0.75\, R_{\mathrm{pass}}$ and $1.10\, R_{\mathrm{pass}}$, and widened radial transition width multiplier 1.6.

The blended floor is paired with sufficient trailing-edge sleeve width. In the V5 operating family, the wider sleeve preserves packet gates under refinement and substantially lowers point peaks. The compact handoff layer then broadens the same principle across entry, catch, and release by assigning pressure containment and derivative smoothing to separate smooth windows.

8.3 Packet-Local Radial Support Profile

The packet-local radial support profile modifies $G = \gamma_{ll}$ through a smooth multiplicative factor:

$$G(l,\sigma) = G_{\mathrm{pre}}(l,\sigma)\exp(g_c W_c(l,\sigma) + g_r W_r(l,\sigma) + g_s W_s(l,\sigma)), \tag{33}$$

where G_{pre} denotes the radial metric before packet-local radial support, W_c is a filled packet/support core window, W_r is a narrow annular catch ring, and W_s is a weaker annular skirt. In the representative

V5 setting,

$$g_c = +0.05, \qquad W_c: \text{ entry/catch/release schedule, radius } 1.3R_{\text{pass}}, \text{ width } 1.8w_{\text{pass}}, \tag{34}$$

$$g_r = +0.12, \qquad W_r: \text{ catch-only annulus, outer radius } 2.4R_{\text{pass}}, \text{ width } 2.4w_{\text{pass}}, \tag{35}$$

$$g_s = +0.05, \qquad W_s: \text{ catch-only annulus, inner/outer radii } 2.4R_{\text{pass}}, 2.6R_{\text{pass}}, \text{ width } 3.2w_{\text{pass}}. \tag{36}$$

The core/ring/skirt structure separates two radial support functions. The narrow ring provides the principal radial-null and angular-pressure localization benefit. The weak skirt provides a smaller smoothing correction to live p_l and p_Ω while preserving the main point-peak relief. High-resolution tests show that this two-feature radial profile keeps all live packet-norm samples safely negative and lowers the radial-null point peak and angular-pressure burden relative to the release-only comparison family.

8.4 Representative Settings

Representative parameter values are listed in Table 2. The values define the working design point and its comparator settings.

Table 2: Representative active-rail parameters and functional implementation decisions.

Parameter	**Representative value or status**
Primary service factor	$V = 5$
Diagnostic low-service factor	$V = 2$ for mechanism checks only
Edge-load service factor	$V = 10$ as edge-load stress setting
Packet radius	$R_{\text{pass}} = 0.35$
Support radius	$R_{\text{th}} = 1.75$
Angular jacket radius	$R_\Omega = 1.75$
Radial support width	$w_{\text{th}} = 0.569$
Packet transition width	$w_{\text{pass}} = 0.06$
Angular jacket width and amplitude	$w_\Omega = 1.4$, $a_\Omega = 0.20$
Lapse cushion strength	$\eta_N = 2.00$
Catch/rematch profile	matched-hold release choreography with minimum-jerk beta fade
Reduced support-shell routing overlay	$a_\beta = +10^{-7}$ as the small reduced-routing target; high-amplitude release-choreography comparators use service shell amplitude 0.5
Support-shell radial band	$0.65\,R_{\text{th}} \le \lvert l \rvert \le 1.20\,R_{\text{th}}$
Support-shell radial shape status	raised-cosine annulus as the cleanest matched-strength support-shell source comparator; release-choreography accounting uses the smooth-box support-shell infrastructure knob
Selected 4D support-shell strength	$\Delta\beta_{\text{sh,max}} \approx 0.20$–$0.25$ for support-shell source comparators
Selected 4D timing band	$\lambda_c = 1.45$–1.55, $\tau_c = 0.30$–0.35
Clock-lapse partner	$\kappa_\alpha / a_\beta \approx 0.375$ for the selected V5 support-shell family
Rail-stretch partner	$\kappa_G / a_\beta = 0$ by default; positive rail-stretch as a comparator for peak shaping

Parameter	Representative value or status
Throat-capacity partner	$\kappa_Q/a_\beta = 0$ by default in the selected support-shell setting; alternate timing and separate intrinsic windows remain comparator knobs
Packet carve/lapse support	packet exclusion 0.90, annular shoulder 0.18, packet-lapse log gain 1.0 with entry/catch/release timing
Packet beta-rematch sleeve	trailing-edge shape, gain 1.8, smooth blended floor 0.60, inner/outer radii $0.75\,R_{\mathrm{pass}}$ and $1.10\,R_{\mathrm{pass}}$; the conservative finite-bundle carrier lead uses sleeve transition multiplier 6.0 and temporal multiplier 1.5, with multiplier 8.0 and temporal multiplier 2.0 as the wider optical-relief bracket
Packet-local radial support profile	filled core gain +0.05, annular catch ring gain +0.12 at radius/width 2.4/2.4, and weak annular skirt gain +0.05 with inner/outer/width 2.4/2.6/3.2
Compact handoff layer	seventh-order smoothstep entry/catch/edge windows, entry service gate near $\sigma = -1.40$, broad edge transition multiplier near 7.2, and separated entry pressure containment versus catch/edge derivative smoothing
Semilocal null-accumulation diagnostic	Gaussian or equivalent smooth radial-null averaging on selected live, handoff, endpoint, and source-sector windows, with affine-reparameterized radial-null benchmark comparisons used as acceptance diagnostics
Source-family grammar	$S_0 + J + R + C_{\mathrm{live}} + G + D_H$, assigning the non-live radial body to a constant-flux string-cloud backbone, endpoint residuals to a coupled junction layer, live burden to handoff trim, angular burden to capacity support, and non-live current to relaxation sectors
Representative endpoint receiver	beta075 $p003\_mid$ negative-l beta-memory receiver with localized angular endpoint support and dense live-clean source-sector accounting
Beta100 timing anchor	beta100 stress operation uses the same receiver grammar with service-specific retiming; the post-release 1.5 anchor gives positive selected, current, and angular transfer accounting on the stress grid
Causal-carrier acceptance	measured GZ-like branch structure is characterized in the shell/throat carrier band and accepted through zero service/carry entry-to-packet reachability, branch-band radial escape, scheduled ADM probes, dense finite-bundle carrier transport, exterior-null service-time rating, and reset-domain wake-tail monitors
Service-time advantage rating	first $s = 15$ exterior-null comparison gives prepared-service advantage ratios about 2.568962 under the schedule-factor proxy and about 1.233312 under the packet-coordinate proxy

Parameter	Representative value or status
Representative V5 status	high-resolution packet-safe, with structured burden separated between live entry/handoff rows, semilocal radial-null exposure, non-live catch/rematch radial-null rows, mitigated CHL-adjacent branch-band optics, favorable exterior-null service-time rating, and beta075 endpoint source-medium closure at finite-difference level

9. SERVICE SEQUENCE

A representative service cycle proceeds through the following phases:

$$\begin{aligned}&\text{prepared substrate} \to \text{ready support plant} \to \text{packet entry} \to \text{locked catch/rematch}\\ &\to \text{support-contained carrying-flow handoff} \to \text{packet release} \to \text{infrastructure reset.}\end{aligned} \tag{37}$$

Passenger-facing accounting is assigned to packet entry, catch/rematch, carried service, release, and selected live buffer windows. System-level ledgers may include reset and decompression through an explicitly selected infrastructure diagnostic window.

10. ADM TRANSLATION

The metric of Eq. (13) is expressed in ADM form through

$$\gamma_{ij} = \operatorname{diag}\left(G, Q, Q\sin^2\theta\right), \qquad \beta^i = (\beta, 0, 0), \qquad \beta_i = (G\beta, 0, 0). \tag{38}$$

The unit normal to a $\sigma = \text{constant}$ slice is

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta}{\alpha}, 0, 0\right), \qquad n_\mu = (-\alpha, 0, 0, 0). \tag{39}$$

Using dot notation for ∂_σ and prime notation for ∂_l, and adopting the convention

$$K_{ij} = \frac{1}{2\alpha}\left(D_i\beta_j + D_j\beta_i - \partial_\sigma\gamma_{ij}\right), \tag{40}$$

the independent extrinsic-curvature components are

$$K_{ll} = \frac{1}{2\alpha}\left(2G\beta' + G'\beta - \dot{G}\right), \tag{41}$$

$$K_{\theta\theta} = \frac{1}{2\alpha}\left(\beta Q' - \dot{Q}\right), \tag{42}$$

$$K_{\phi\phi} = K_{\theta\theta}\sin^2\theta. \tag{43}$$

The mixed eigenvalue-like quantities are

$$k_l = K^l{}_l = \frac{1}{2\alpha}\left(2\beta' + \beta\frac{G'}{G} - \frac{\dot{G}}{G}\right), \tag{44}$$

$$k_\Omega = K^\theta{}_\theta = K^\phi{}_\phi = \frac{1}{2\alpha}\left(\beta\frac{Q'}{Q} - \frac{\dot{Q}}{Q}\right). \tag{45}$$

Thus

$$K = k_l + 2k_\Omega, \qquad K_{ij}K^{ij} = k_l^2 + 2k_\Omega^2. \tag{46}$$

For spatial line element $\mathrm{d}l_3^2 = G\mathrm{d}l^2 + Q\mathrm{d}\Omega^2$, the scalar curvature is

$$^{(3)}R = -\frac{2Q''}{GQ} + \frac{(Q')^2}{2GQ^2} + \frac{G'Q'}{G^2Q} + \frac{2}{Q}. \tag{47}$$

The Hamiltonian and radial momentum constraints produce the ADM ledger fields

$$16\pi\rho_{\mathrm{ADM}} = {}^{(3)}R + K^2 - K_{ij}K^{ij} = {}^{(3)}R + 4k_lk_\Omega + 2k_\Omega^2, \tag{48}$$

$$8\pi j_l = -2k'_\Omega + \frac{Q'}{Q}(k_l - k_\Omega), \tag{49}$$

$$j_\theta = 0, \qquad j_\phi = 0. \tag{50}$$

11. SOURCE-PLACEMENT LEDGER

The source-placement ledger assigns demanded source roles to system components. The ledger is used for engineering localization, source-family accounting, and representative-design acceptance.

Source role	**Engineering assignment**	**Primary diagnostic**
Standing support substrate	Prepared throat/core, support shell, baseline clock-lapse, and angular jacket	baseline ρ_{ADM} and pressure channels
Radial support edge	Radial softening and support-shell gradient control	p_l, edge gradients
Throat-capacity/angular jacket	Angular and throat support capacity	p_Ω, T_{kk} side effects
Clock-lapse cushion	Causal-margin and timing support	packet norm margin, T_{kk} response
Carrying-flow field	Support-contained scheduled service flow	Δj_l, packet norm
Support-shell overlay	Annular support infrastructure timing component	catch/support Δj_l fraction, 4D source ratios
Catch/rematch absorber	Inner packet-edge handoff absorber	Δj_l, T_{kk} residual
Release choreography	Matched hold, derivative-limited beta fade, and coordinated lapse/carve relaxation	packet norm margin, release derivative maxima, live high-burden channel exposure
Compact handoff layer	Endpoint-smooth entry containment, catch bearing, and edge smoothing	live p_l, j_l, p_Ω, semilocal T_{kk}
Entry service gate	Service boundary that separates setup support from live packet scoring	live packet norm, live source-role closure
Trailing-edge beta-rematch sleeve	Smooth packet-edge correction for local $(v_{\mathrm{coord}} + \beta)$ mismatch and branch-band carrier-collar width preservation	packet-norm safety, live T_{kk} fraction, point peaks, dense bundle width

Packet-local radial support profile	Mild γ_{ll} core stretch plus annular catch ring and weak skirt	T_{kk}, p_l, p_Ω, radial metric derivative diagnostics
Semilocal null-accumulation ledger	Finite-window radial-null averaging across live packet, handoff, endpoint, and source-sector regions	finite-window T_{kk} floor margin and negative-part integral
Constant-flux radial string-cloud backbone	Non-live radial-tension support scaffold for the dominant A/B/I radial infrastructure sector	ρ, p_l, areal flux proxy, live fraction
Endpoint/junction layer	Coupled support-edge and reset/decompression endpoint source target carrying residual radial trim, angular support, and current relaxation	selected-null deficit, current, p_Ω, finite-width closure, regulated medium response
Beta-memory endpoint receiver	Negative-l support-edge receiver coupled to beta-release transfer memory	endpoint transfer accounting, live gate, source-sector closure
Endpoint current regulator and medium	Non-live regulated anisotropic heat/current endpoint medium with a small radial-pair regulator and internal angular response	radial-block class, heat-flux margin, angular closure, live exclusion
Causal carrier layer	Controlled one-way radial carrier structure generated by α, β, and γ_{ll}	branch margins, entry reachability, radial escape, probe evolution
Reset/decompression infrastructure	Post-service restoration of support state	infrastructure ledger

The reduced ADM diagnostics indicate that the selected small support-shell overlay places its incremental j_l demand overwhelmingly in catch/rematch support infrastructure while keeping packet $\Delta\rho$ effectively quiet. The 4D demanded-source diagnostics sharpen that result: the same component is packet-safe and controllable as a continuous metric overlay, while larger single-channel carrying-flow amplitude identifies the radial-null and current channels assigned to coupled support. The release-choreography, radial-support, compact-handoff, and entry-gate ledgers further sharpen the design. V5 packet safety is maintained through a smoother blended trailing-edge beta rematch, two-feature packet-local radial support, compact pressure-aware entry/catch/edge shaping, and live scoring that begins inside the supported entry corridor. The demanded-source burden is structured: live angular/current peaks are assigned to packet-in-support rows, semilocal radial-null exposure is measured by finite-window null-accumulation diagnostics, non-live radial-null plus radial-pressure burdens are carried by core-throat, support-edge, and endpoint infrastructure, and the frozen beta075 endpoint package is represented by a regulated non-live heat/current medium rather than by an unconstrained interpolation source.

11.1 Semilocal Null-Accumulation Source Ledger

The source-placement ledger includes semilocal radial-null accounting for comparing pointwise T_{kk} concentration with finite-window exposure. The diagnostic follows quantum-motivated finite-window null-energy logic in a prescribed-metric engineering ledger [7]. For a diagnostic center (l_c, σ_c), define

$$\langle T_{kk} \rangle_{W_{\mathrm{SN}}} = \frac{\sum_i W_{\mathrm{SN}}(l_i, \sigma_i; l_c, \sigma_c)\, T_{kk}(l_i, \sigma_i)}{\sum_i W_{\mathrm{SN}}(l_i, \sigma_i; l_c, \sigma_c)}, \tag{51}$$

where W_{SN} is a compact Gaussian or equivalent smooth averaging window on the selected live-packet, handoff, endpoint, or source-sector scope. A representative benchmark compares this finite-window value with

$$T_{kk,\mathrm{floor}} = -\frac{8\pi B}{\tau^2}, \qquad B = \frac{1}{32\pi}, \tag{52}$$

so that $\tau = 1$ gives $T_{kk,\mathrm{floor}} = -0.25$ in ledger units. The compact handoff family improves the finite-window radial-null margin on the live catch/rematch packet window while also reducing broad angular-pressure exposure. The dense entry-gated source-sector closure, intermediate source targets, and endpoint/junction receiver ledgers retain positive benchmark margin on the finite windows used for representative acceptance.

For accepted radial-null SNEC accounting, the window coordinate is affine-reparameterized along the sampled radial null branch. With coordinate-time tangent

$$K^\mu = (1, v_\pm), \tag{53}$$

the diagnostic evaluates

$$K^\nu \nabla_\nu K^\mu = \kappa K^\mu + \mathcal{R}^\mu_\perp, \tag{54}$$

where κ is the measured non-affinity and $\mathcal{R}^\mu_\perp$ is the remaining radial geodesic residual monitor. The affine weight is integrated from κ and normalized at the window center. In the repaired beta075 *rematch_w6_t1p5* package, the full-stride affine-reparameterized radial-null screen evaluates 137,064 windows at affine widths 0.5, 1.0, and 2.0 in both component-sector and intermediate residual-inclusive modes, with zero all-window and zero scoreable benchmark-floor violations. The tightest scoreable broad-window margin is approximately 0.051880 and remains non-live. This diagnostic expresses source cost at the finite engineering scale of the packet, handoff, endpoint, and source-sector structures and complements the point-peak ledger.

12. CATCH-EDGE CONTROL LAW

Let

$$\xi = l - s(\sigma) \tag{55}$$

be a packet-centered radial coordinate. A preferred absorber amplitude is a compact kinematic handoff law multiplied by an inner-edge envelope:

$$A_{\mathrm{catch}}(\xi,\sigma) = W_{\mathrm{catch}}(\sigma)\left[a\, H_{\mathrm{flow}}(\xi,\sigma) + b\, B_{\mathrm{inner}}(\xi)\right], \tag{56}$$

where W_{catch} is a locked timing window, H_{flow} is a smooth carrying-flow handoff or flow-mismatch measure, and B_{inner} is a compact bump centered near $\xi = -R_{\mathrm{pass}}$. A residual-responsive limiter may bound the amplitude:

$$A_{\mathrm{catch}} \le C\left([R_{T_{kk}}]_+\right), \tag{57}$$

where C denotes a smooth cap and $R_{T_{kk}}$ denotes the predicted radial-null residual. The primary control law is kinematic; the residual-responsive term supplies protective limiting and diagnostic alignment.

13. STANDING SUBSTRATE AND SOURCE-LEDGER RESULTS

Standing substrate subtraction isolates service-induced ADM demand from prepared infrastructure demand. For a live service field F_{live} and a time-matched prepared carrying-flow-off substrate field F_{sub}, define

$$\Delta\rho_{\mathrm{ADM}} = \rho_{\mathrm{ADM,live}} - \rho_{\mathrm{ADM,sub}}, \tag{58}$$

$$\Delta j_l = j_{l,\mathrm{live}} - j_{l,\mathrm{sub}}. \tag{59}$$

This diagnostic separates baseline support curvature from active service increments. The heavier demanded-source ledger evaluates

$$T_{\mu\nu}^{\mathrm{dem}} = \frac{1}{8\pi} G_{\mu\nu}[g] \tag{60}$$

from the prescribed four-dimensional metric and records radial null demand, radial pressure, angular pressure, radial current, and packet-comoving density.

Table 4: Representative reduced ADM and 4D demanded-source evidence.

Diagnostic layer	**Representative result**
$V = 5$ exact ADM and substrate subtraction	Live packet norm samples are safe. The packet contribution to subtracted $\Delta\rho$ is about 0.000295% of live-packet ρ, while live-packet Δj_l is about 11.75% of live-packet j_l.
$V = 10$ reduced ADM edge-load comparison	Subtracted $\Delta\rho$ is small in the reduced-support target. The live-packet Δj_l fraction rises to about 19.25%, with about 97.94% of live-packet Δj_l localized in catch/rematch. This supplies the edge-load stress rating for the same architecture.
Reduced support-shell routing target	At $V = 5$, the selected $a_\beta = +10^{-7}$ target gives catch/support incremental Δj_l fraction 0.9928425, packet incremental Δj_l fraction 3.19×10^{-7}, packet incremental $\Delta\rho = 5.44 \times 10^{-16}$, and zero packet norm violations.
Small-target $V = 10$ packet safety	The same small reduced support-shell target is packet-safe at the edge-load setting and preserves catch/support localization above 0.9927. The high-amplitude release-choreography family is rated separately at the V5 operating point.
Single-channel continuous carrying-flow overlay	High-amplitude continuous carrying-flow overlays are packet-safe at $V = 5$ and identify radial-null and current burden channels. At $a_\beta = +0.5$ on the high-resolution check, max total burden ratio is about 1.090805, radial-null ratio about 1.056153, and radial-current ratio about 1.040020.
Timed coupled support-shell refinement	High-resolution timing checks select a delayed, narrower support-shell event. With positive carrying-flow overlay, $\lambda_c = 1.45$–1.55, $\tau_c = 0.30$, $\kappa_\alpha/a_\beta = 0.375$–$0.5$, and $\kappa_G/a_\beta = 0$, max total burden improves to about 1.053–1.054 in the smooth-box support-shell family.
Throat-capacity accounting	Same-window throat-capacity coupling gives its best source-objective and aggregate behavior at $\kappa_Q/a_\beta = 0$ in the V5 support-shell setting. Negative ratios offer small localized tradeoffs with peak and radial-current penalties, while positive ratios raise radial-null burden.
Matched-strength shape selection	At equal peak $\Delta\beta_{\mathrm{sh,max}} = 0.25$, the raised-cosine annulus outperforms smooth-box and Gaussian annular profiles. The best matched-strength case gives source-objective score 0.803284, max burden ratio 1.013261, live packet-norm drift 3.09×10^{-8}, radial-null ratio 1.007942, radial-current ratio 1.002684, and angular-pressure ratio 1.013261.

Diagnostic layer	Representative result
Strength robustness ladder	The selected raised-cosine annulus is packet-quiet over $\Delta\beta_{\mathrm{sh,max}} =$ 0.15–0.35. The preferred operating band is 0.20–0.25; higher targets are useful edge-stress cases for point-peak and shell-throat concentration diagnostics.
Release choreography	A V5 matched-hold release law with minimum-jerk beta fade, packet carve/lapse support, and a trailing-edge beta rematch preserves packet safety on the representative grid with all live packet-norm samples safely negative and zero high-burden live top points.
Blended trailing-edge beta rematch	A smooth blended packet-center floor with a widened trailing-edge sleeve survives 81×109 with all live packet-norm samples safely negative, zero high-burden live top points, max packet norm about -5.455516, live T_{kk} fraction about 0.024255, live p_l fraction about 0.006986, max total burden ratio about 2.035942, and max point peak about 8.627056. The finite-bundle carrier lead widens the same trailing-edge rematch family for branch-band bundle preservation.
Two-feature radial support profile	Adding a mild γ_{ll} core stretch, narrow annular catch ring, and weak annular skirt preserves packet safety and keeps the main point-peak relief. At high resolution it lowers the radial-null point peak by about 4.0% and lowers live angular-pressure burden by about 2.45% relative to the corresponding blended release ledger.
V5 algebra ledger	On the full-range 81×109 algebra ledger, live packet safety is hard-bounded with zero positive live packet-norm points and max live packet norm about -5.455491. Top live angular/current rows lie in entry-precatch packet-in-support samples ahead of the radial core/ring/skirt window onset, while top non-live radial-null rows lie in catch/rematch core-throat and support-edge samples.
Compact handoff law	At 61×83, the compact wide-edge handoff lowers live T_{kk} by about 10.8%, live j_l by about 10.2%, and live p_Ω by about 35.9% relative to the split-carve reference while preserving packet safety. The same compact handoff reduces channel-cause radial-metric, lapse-curvature, beta-gradient, and radial second-derivative scores in the live-badness ledger.
Semilocal compact handoff	On the live catch/rematch packet scope with $\tau = 1$, the split-carve reference gives finite-window $T_{kk,\min} \approx -0.2317$ against the -0.25 benchmark floor. The compact wide-edge handoff gives finite-window $T_{kk,\min} \approx -0.1974$, increasing the finite-window margin from about 0.0183 to about 0.0526 while reducing the negative-part integral.
Entry service gate and compact handoff	On the extended 101×151 domain, the representative entry-gated compact handoff has zero positive live packet-norm samples and max live packet norm about -1.2424. Live channel assignment closes with zero live residual fraction for radial-null, radial-pressure, radial-current, and angular-pressure burdens.

Diagnostic layer	Representative result
Pressure-aware entry/handoff shaping	Stronger broad entry containment lowers live T_{kk} and live p_l, while separated entry, catch, and edge functions assign angular/current and point-peak channels to distinct source roles. The representative pressure-aware handoff separates entry pressure containment from catch/edge derivative smoothing so those channels can be tuned independently.
Full-grid source decomposition	The entry-gated compact wide4 source picture is support-plant dominated. The integrated live fractions are about 2.889% for radial-null burden, 0.787% for radial pressure, 14.706% for angular pressure, and 11.976% for current, with zero positive live packet-norm samples.
Dense source-sector closure	On the dense entry-gated sector closure, constrained source sectors assign about 98.659% of radial-null burden, 98.716% of radial-pressure burden, and 91.315% of angular-pressure burden into explicit source roles. The promoted non-live current-relaxation sector carries the remaining small current burden with zero live burden.
Source-family target split	The principal source target is the A/B/I non-live radial infrastructure family. In representative source-target accounting, A/B/I carries about 94.03% of the target burden, live handoff trim about 2.88%, angular capacity about 1.50%, and current control about 0.11%.
Constant-flux radial string-cloud backbone	The A/B/I body is close to an NEC-saturating radial-tension scaffold with $p_l/\rho \approx -0.999288$, zero live scaffold fraction, and selected-null deficit about 1.338% of the string-like scaffold. The areal string-flux proxy has weighted relative mean deviation about 0.0904%.
Endpoint-J source freeze	After subtracting the constant-flux radial backbone, the residual endpoint source is frozen as a bounded reset-decompression cap body plus a finite support-edge closure component. The package remains non-live on baseline and dense ledgers; the dense source regeneration has 90,857 rows, zero positive live packet-norm samples, and max live packet norm about -6.333723. The dense support-edge closure remains tiny, with max closure coefficient about 0.000846 in the dedicated dense stability rung.
Beta-memory endpoint receiver	The representative effective receiver grammar uses a beta-memory negative-l endpoint receiver. The beta075 $p003\_mid$ receiver is live-clean in dense promotion and full-stride source-sector checks. The beta100 release-timing stress condition uses the same grammar with service-specific retiming as a stress anchor.
Endpoint source class and current regulator	The frozen endpoint source is not scalar-only and is not fully represented by an ordinary Type-I anisotropic fluid. Its first viable physical class is a regulated anisotropic heat/current medium. A small non-live $\rho + p_l$ regulator removes the flux-dominant radial-block rows with regulator/source burden about 0.037723 on the baseline mesh and 0.039030 on the dense mesh, while preserving zero live regulator rows.

Diagnostic layer	**Representative result**
Regulated endpoint medium closure	At the 1.10 regulator safety factor, the regulated medium has no post-regulator Type-IV rows, no superluminal or undefined boost rows, and regulator/source burden about 0.041495 on the baseline mesh and 0.042933 on the dense mesh. The constrained field-closure validation carries the non-live angular-pressure watch as an internal medium response: worst normalized angular L1 error is about 0.002977 on the baseline mesh and 0.009524 on the dense mesh, with zero live, regulator-live, and superluminal rows.
Affine-reparameterized null-accumulation accounting	Semilocal radial-null windows are applied to live, handoff, endpoint, and source-sector scopes using an explicitly reparameterized affine window coordinate. The repaired beta075 $rematch\_w6\_t1p5$ package has zero all-window and zero scoreable benchmark-floor violations across 137,064 full-stride windows at affine widths 0.5, 1.0, and 2.0 in both component-sector and intermediate residual-inclusive modes.
Causal-carrier reachability	The shell/throat carrier band contains a measured GZ-like cone signature with measured branch-crossing structure. Entry-side causal floods from boundary and support-edge seed sets record zero service-stage and carry-stage entry-to-live-packet hits in the sampled ledgers.
Branch-band radial escape	In the promoted beta075 receiver setting, live packet, packet geometry, main carrier, support plant, and live branch-band seeds reach exterior radial boundaries on both radial branches. Extended $s = 12$ and $s = 15$ confidence domains give radial escape for all 1056 core-mask seed120 traces while the local branch-crossing diagnostic remains present.
Dense finite-bundle carrier transport	In the tensor-consistent beta-collar generator setting, the conservative $rematch\_w6\_t1p5$ lead preserves 136/136 dense-ray radial escape, 136/136 recovery from both-shrinking intervals, zero sustained-to-end both-shrinking rays, and 0/8 caustic-like bundles on the $s = 15$ confidence domain. The worst all-both l-width ratio is about 0.275792, compared with about 0.009603 in the corresponding unmitigated dense-bundle carrier run. A wider $rematch\_w8\_t2p0$ bracket also gives 0/8 caustic-like bundles and worst all-both l-width about 0.394689.
Scheduled ADM probe evolution	Prescribed-ADM probes through $\alpha, \beta, \gamma_{ll}, \gamma_{\Omega\Omega}$ preserve radial, off-axis, and small-bundle branch-band escape. Reset-decompression wake-tail traces remain exterior to live service and record zero live re-entry in the confidence run.

Diagnostic layer	Representative result
Service-time advantage ledger	The first $s = 15$ exterior-null service-time ledger rates the prescribed schedule as operationally favorable. Prepared service time is about 3.045; the schedule-factor A-to-B proxy gives distance about 7.822491 and advantage ratio about 2.568962, while the packet-coordinate proxy gives distance about 3.755436 and advantage ratio about 1.233312. With request-triggered setup included in the service clock, the ratios remain favorable at about 2.487278 and 1.194097.
V10 edge-load release rating	The high-amplitude release family supplies an edge-load stress condition for the same architecture. The representative V5 operating embodiment is rated on V5 packet safety, source localization, compact handoff behavior, and semilocal finite-window null margin.

The combined evidence defines the design interpretation. The small support-shell overlay is a reliable reduced-harness support-routing component. The larger 4D source-relief objective uses coupled metric timing, radial gradient matching, derivative-smooth release choreography, compact pressure-aware handoff shaping, semilocal null-accumulation accounting, packet-local radial support placement, and explicit source-family assignment. Clock-lapse timing is the strongest default partner for aggregate burden reduction, raised-cosine annular support is an important matched-strength comparator, and same-window throat-capacity is reserved for alternate intrinsic-window accounting. The representative V5 end-to-end operating embodiment is the blended trailing-edge release-choreography family with finite-bundle beta-collar widening, two-feature packet-local radial support, entry service gating, compact entry/catch/edge handoff support, dense source-sector closure, beta075 *p003_mid* endpoint receiver accounting, affine-reparameterized radial-null margin, causal-carrier gates, and exterior-null service-time rating. It is packet-safe under smoothing and refinement, preserves dense branch-band bundle transport in the carrier-collar audit, records favorable A-to-B service-time advantage, and localizes demanded-source burden into separately diagnosed live handoff trim, non-live radial string-cloud support, coupled endpoint/junction structure, angular capacity, current relaxation, and reset-domain wake-tail restoration. The endpoint source realization is no longer only a target grammar: for the repaired beta075 lead it is a finite non-live source package with a regulated anisotropic heat/current medium that passes constrained finite-difference field-closure diagnostics.

14. SOURCE-FAMILY TARGET ARCHITECTURE

The source-family target architecture translates the demanded-source ledger into source roles and endpoint-medium closure tests. It retains the prescribed metric as the geometric service program and assigns demanded stress into a compact set of source targets. Non-minimally coupled scalar and traversable-wormhole source constructions provide useful admissibility references for energy-condition violation, averaged null energy, and trans-Planckian source thresholds [6]. The target grammar is

$$\mathcal{S}_{\text{target}} = S_0 + J + R + C_{\text{live}} + G + D_H, \tag{61}$$

where S_0 is the constant-flux radial string-cloud backbone, J is the coupled endpoint/junction layer, R is small core-body residual leakage, C_{live} is live handoff trim, G is angular-capacity support, and D_H is non-live current relaxation. This grammar separates large non-live infrastructure support from packet-facing handoff trim and from endpoint closure, while keeping all source roles attached to explicit metric-service components.

Target sector	Assigned source role	Representative diagnostic
S_0 radial string-cloud backbone	Non-live radial-tension scaffold for the dominant A/B/I support infrastructure	$\rho > 0$, $p_l \approx -\rho$, constant areal flux
J endpoint/junction layer	Coupled support-edge shoulder and reset/decompression cap carrying radial trim, angular capacity, and current relaxation	selected-null deficit, current, p_Ω, finite closure
R core-body residual leakage	Small residual core-body source after subtracting the radial backbone	residual selected-null and pressure leakage
$C_{\rm live}$ live handoff trim	Packet-facing handoff trim for angular/current, radial-null, and radial-pressure service rows	live fractions, packet norm, handoff localization
G angular-capacity support	Non-live angular and throat-capacity support outside the packet corridor	p_Ω support and shell/throat placement
D_H current relaxation	Non-live reset/background and distributed current relaxation	j_l, transfer accounting, conservation closure

The dominant non-live radial support has the local algebra of a radial-tension scaffold. A representative unit-frame fit gives

$$\frac{p_l}{\rho} \approx -0.999288, \qquad \frac{j_l}{\rho} \approx -0.000485, \qquad \frac{p_\Omega}{\rho} \approx 0.001078, \tag{62}$$

with zero live scaffold fraction in the source-target ledger. The areal flux proxy

$$\mu = \frac{\Phi}{\gamma_{\Omega\Omega}} \tag{63}$$

is nearly constant across the principal radial body, giving the radial backbone a concrete string-cloud realization target. In the representative dense source-family target, A/B/I carries about 94.03% of the target burden; the selected-null deficit after subtracting the string-like scaffold is about 1.338% of the scaffold, has zero live fraction, and is concentrated in non-live support-edge and reset/decompression structures. The areal string-flux proxy has weighted relative mean deviation about 0.0904%, giving the principal radial body the constant-flux radial string-cloud interpretation.

The endpoint/junction layer is the primary coupled realization target after the radial backbone. It combines the support-edge shoulder and reset/decompression cap into one source role because the same endpoint region carries residual radial trim, angular endpoint support, and current relaxation. In representative junction accounting, the coupled J layer carries selected-null deficit about 1.010455, current about 0.381351, and angular support about 3.961788, with zero live rows and zero live selected-null burden.

In the repaired beta075 design, J is represented by a frozen finite endpoint source package. The reset-decompression cap is a bounded no-tail structured body, and the support edge is a finite closure component rather than a singular shoulder. On the dense 377×241 source-ledger refinement the endpoint source remains non-live, the reset-cap coefficients remain bounded, and the support-edge closure stays tiny and finite-width. This is the effective-source plant carried forward by the disclosure.

The physical-source class of the frozen endpoint plant is a regulated anisotropic heat/current medium. Scalar-only descriptions are algebraically too narrow, and an ordinary Type-I anisotropic fluid alone does not cover the flux-dominant radial-block rows. A small non-live regulator in $\rho+p_l$ makes the radial heat/current block boost-diagonalizable while leaving $\delta j_l = 0$ and adding no angular, collar, packet, or extra closure component. At the selected 1.10 regulator safety factor, the regulator/source burden is about 0.041495 on the baseline mesh and 0.042933 on the dense mesh, with positive transport margin and zero live regulator rows.

The same regulated medium carries the non-live angular-pressure watch as an internal anisotropic response. The constrained field-closure validation preserves the frozen reset-cap body, finite support-edge closure, and endpoint current regulator, and allows only this internal angular response. It closes the angular response on both baseline and dense meshes with worst normalized angular L1 errors about 0.002977 and 0.009524, respectively, with zero live rows, zero regulator-live rows, and zero superluminal or undefined boost rows. Thus the beta075 endpoint source has a first constructive physical-source candidate at finite-difference field-closure level. The claim remains bounded: this is not a covariant matter-action theorem, a semiclassical/RSET proof, or a global horizon theorem.

The beta-memory endpoint receiver supplies the effective receiver mechanism within this grammar. Its amplitude is tied to accumulated beta-release transfer and applied through a negative-l support-edge receiver channel. The representative beta075 $p003\_mid$ receiver provides the promoted effective receiver setting for the V5 source-target architecture: it is live-clean, endpoint-effective, and positive-margin in the full-stride affine-reparameterized radial-null package. The beta100 release-timing stress condition uses the same grammar with service-specific receiver retiming; the post-release 1.5 anchor gives positive selected, current, and angular transfer accounting while preserving the live gate and SNEC companion margin.

15. CAUSAL CARRIER AND SHELL-THROAT CORRESPONDENCE DIAGNOSTIC

Garattini and Zatrimaylov describe a wormhole–warp-drive correspondence in which a Morris–Thorne wormhole metric can be written in a warp-drive-like form by allowing nonzero intrinsic spatial curvature in the warp-drive metric [5]. Their correspondence identifies a shell/throat layer where an interpolation profile, a shift-like sector, intrinsic metric curvature, and intrinsic metric derivatives enter the same curvature expressions. In the active-rail notation, the support-shell timing window W_{sh} supplies the localized shell/interpolation profile, β is the carrying-flow or shift-like sector, and G, Q are the intrinsic support/throat metric components. The active-rail carrier band therefore lies in the nearby design space where wormhole/throat structure and warp-shift structure meet: local radial cones can tilt far enough that one coordinate null branch crosses zero while the stationary-vector indicator changes sign in the same active support region.

Clark, Hiscock, and Larson studied null geodesics in the Alcubierre warp-drive spacetime and found horizon-like conical access structures for superluminal warp-bubble operation [1]. Later geodesic studies of the Alcubierre metric likewise treat null and timelike geodesic behavior as a central optical diagnostic for warp-shift spacetimes [2]. The active rail is a throat-loaded service-carrier architecture, and the CHL comparison supplies a nearby optical diagnostic reference. The comparison is CHL-adjacent: it supplies the relevant diagnostic lesson that shift-like service geometry is evaluated most cleanly at both the pointwise/single-ray scale and the finite-bundle scale. GZ is the closer reference for throat/branch correspondence; CHL is the closer reference for null-geodesic access, cone formation, and finite-bundle optical behavior. The active-rail carrier audit therefore evaluates both the local branch structure and the dense bundle transport through that branch structure.

The superluminal timing comparison supplies a separate chronology-governance requirement. Everett–Roman tube networks and Shoshany–Snodgrass two-warp constructions show that superluminal service links acquire closed-timelike-curve relevance when multiple service legs are composed across incompatible endpoint orderings [3, 4]. The active rail treats that literature as a safety-engineering constraint on service sequencing: the transport channel is armed by a rail-time governor, reset and endpoint-recovery windows are scheduled as service-readiness intervals, and connected rail networks require one shared chronology policy.

A single-channel overlay

$$\beta = \beta_0 + a_\beta W_{\rm sh} \tag{64}$$

therefore supplies a shell-like carrying-flow feature while assigning intrinsic metric response to the G and Q partners. The demanded-source ledger detects burden where $W_{\rm sh}$ overlaps gradients in G and Q. The raised-cosine annular bearing shell, delayed support timing, clock-lapse coupling, and radial-support placement treat this overlap as a coordinated carrier layer with timing and gradient matching across shell/throat structure.

The causal carrier diagnostic evaluates the radial null speeds

$$v_\pm(l,\sigma) = -\beta(l,\sigma) \pm \frac{\alpha(l,\sigma)}{\sqrt{\gamma_{ll}(l,\sigma)}}, \tag{65}$$

the stationary-vector indicator

$$g_{\sigma\sigma} = -\alpha^2 + \gamma_{ll}\beta^2, \tag{66}$$

branch-margin maps, shell/throat-overlap scores, entry reachability, radial escape, prescribed-ADM probe evolution, dense finite-bundle carrier transport, and exterior-null service-time rating. This diagnostic measures the active rail as a controlled causal carrier: local one-way branch structure can be present in the carrier band, while the live packet and main carrier are accepted by service-stage reachability, radial escape, finite-bundle transport, and operational A-to-B timing gates.

In the representative beta075 receiver setting, the local branch-crossing structure is measured directly alongside packet norm and source-channel ledgers. The refined beta075 branch band has a sampled live-packet minimum branch margin of approximately 0.002402, with live-packet branch-crossing edges and nonnegative $g_{\sigma\sigma}$ points in the active shell/throat overlap band. This preserves the measured GZ-like cone signature as a real carrier-layer feature and assigns its service consequence to causal reachability and escape gates.

The first paired gate is entry-to-packet reachability. Sampled radial causal floods from domain boundaries, support outer edges, and main-support outer edges record zero service-stage and carry-stage entry-to-live-packet hits across the promoted beta075 receiver, beta100 stress cases, receiver-retimed cases, and causal-margin-guard variants. Post-release main-support-edge hits are assigned to reset/wake restoration accounting, where they are tracked separately from live service.

The second paired gate is live-service radial escape. For the promoted beta075 $p003\_mid$ mechanism on the refined long-tail domain, live packet, packet geometry, main carrier, support plant, and live branch-band seeds reach exterior radial boundaries in both radial branches. Increased targeted seed density preserves that result for the live packet, main carrier, support plant, and branch-band families; the branch-band minus families pass through the branch-sensitive region and reach exterior radial boundaries. On the extended $s = 12$ and $s = 15$ confidence domains, all 1056 core-mask seed120 traces reach radial boundaries while the local branch-crossing diagnostic remains present.

The prescribed-ADM probe audit provides the richer scheduled-geometry read. Evolving probes through $\alpha(\sigma,l)$, $\beta(\sigma,l)$, $\gamma_{ll}(\sigma,l)$, and $\gamma_{\Omega\Omega}(\sigma,l)$ preserves radial, off-axis, and congruence branch-band escape. The finite-bundle carrier audit supplies the stronger acceptance criterion: the branch-band preserves neighboring-ray width and ordering at the carrier scale in addition to individual ray recovery. With the tensor-consistent widened trailing-edge beta-rematch collar, the representative $s = 15$ dense congruence audit gives 136/136 dense-ray radial escape, 136/136 recovery from both-shrinking intervals, zero sustained-to-end both-shrinking rays, and 0/8 caustic-like bundles. The conservative $rematch\_w6\_t1p5$ setting gives worst all-both l-width ratio about 0.275792, and the wider $rematch\_w8\_t2p0$ bracket gives about 0.394689. The exterior-null service-time ledger supplies the operational A-to-B rating: the representative $s = 15$ schedule gives prepared-service advantage ratios about 2.568962 under the schedule-factor proxy and about 1.233312 under the packet-coordinate proxy, with favorable request-triggered accounting as well. The explicit reset-decompression wake-tail monitor clears monotonically with future-domain extension, remains exterior to live service, and records zero live re-entry in the tested confidence run. The remaining wake-tail traces belong to reset-domain tail-closure monitoring.

The resulting acceptance language is componentized. The active rail includes a measured GZ-like cone signature as a carrier-band feature, CHL-adjacent branch-band optics as a finite-bundle diagnostic comparison class, live-service radial escape as a prescribed-metric operating property, mitigated finite-bundle carrier transport as a branch-band acceptance gate, exterior-null service-time advantage as an operational rating, and endpoint/wake tail closure as a reset-domain diagnostic. These diagnostics support a design in which the protected live packet corridor is carried by support infrastructure while causal access, source placement, branch-band transport, endpoint timing, and endpoint restoration are evaluated by separate gates. The disclosure therefore states the current prescribed-metric result positively: the repaired beta-collar carrier passes the tested CHL-adjacent branch-band optics gate with zero sustained trapping traces and zero dense caustic-like bundle-collapse flags. Higher-tier causal-structure and source-family realization diagnostics separately address global-horizon exclusion and physical matter-model scope.

16. DIAGNOSTIC ARCHITECTURE AND ACCEPTANCE PATH

16.1 Reusable Diagnostic Layers

The diagnostic architecture includes the following reusable layers:

1. Exact reduced-field builder and ADM ledger for α, β, G, and Q, including packet norm, region labels, service labels, ρ_{ADM}, and j_l.
2. Standing substrate subtraction using carrying-flow-off or ready-state fields to isolate service-induced $\Delta\rho$ and Δj_l.
3. V5 support-shell ansatz accounting, selecting the direct annular support-shell window over momentum-following controls.
4. Reduced diagnostic ladder for the small support-shell target, including target gates, rich objective scoring, sign comparison, packet-safety overlay, balance bookkeeping, and ramp classification.
5. V10 edge-load checks for the small support-shell target.
6. Continuous metric-side source-ledger characterization of the support-shell carrying-flow overlay.
7. Coupled timing sweeps adding clock-lapse and rail-stretch partners to the continuous support-shell metric expression.
8. Same-window throat-capacity accounting, identifying $\kappa_Q/a_\beta = 0$ as the default same-window throat-capacity setting.
9. Support-shell shape selection, identifying the raised-cosine annulus as the preferred radial bearing shape at matched peak support-shell strength.
10. Matched-strength robustness ladder for the selected raised-cosine annulus support-shell family.
11. Arbitrary service-factor input generation for reduced ADM ladder hardening and interpolation checks.
12. Packet carve/lapse and annular shoulder sweeps, showing that packet safety and live high-burden channel exposure depend on both causal-margin restoration and support-edge shaping.
13. Packet beta-rematch and release-choreography accounting, including matched hold, minimum-jerk beta fade, coordinated lapse/carve support, and shaped packet-edge beta rematch.
14. V5 floor-union refinement accounting for the release family, showing that a smooth blended beta floor plus widened trailing-edge sleeve preserves packet safety at higher grid resolution.
15. Packet-local radial support accounting, identifying a mild filled core stretch, narrow annular catch ring, and weak annular skirt as the representative radial support profile.
16. Algebra ledger, separating live angular/current peaks from non-live catch/rematch radial-null peaks.

17. Compact handoff accounting, identifying endpoint-smooth entry/catch/edge support and the entry service gate as the pressure-aware handoff profile for V5 source placement.
18. Full-grid source decomposition, dense source-sector closure, and constrained source-role assignment for radial support, angular capacity, live handoff trim, and current relaxation.
19. Source-family target extraction, including the constant-flux radial string-cloud backbone, endpoint/junction layer, small residual leakage, angular-capacity jacket, and live handoff trim.
20. Beta-memory endpoint receiver accounting, including release-transfer memory, negative-l receiver placement, endpoint transfer accounting, and service-factor retiming anchors.
21. Endpoint source-medium accounting, including the frozen endpoint-J source package, support-edge closure component, source-class classification, current-regulator budget, regulated-medium admissibility, and constrained field-closure validation.
22. Affine-reparameterized radial-null SNEC and finite-window null-accumulation diagnostics comparing live, handoff, endpoint, and source-sector windows with selected radial-null benchmark floors and residual monitors.
23. Causal carrier diagnostics, including shell/throat overlap maps, branch-margin maps, entry reachability, long-tail radial escape, scheduled-ADM probe evolution, dense finite-bundle branch-band transport, exterior-null service-time rating, and chronology-governed service sequencing.

16.2 Source-Family Boundary

The disclosure establishes the V5 prescribed-metric architecture, demanded-source ledger behavior, source-family target grammar, and beta075 endpoint source-medium closure package. Matter-source realization is organized through source-family accounting and finite-difference source-medium closure diagnostics. A metric family is design-grade for source-family realization when it satisfies the following:

1. live packet norm is safely negative under the selected service factor;
2. packet $\Delta\rho$ and packet-comoving negative-density increments remain quiet;
3. incremental ADM momentum is localized in catch/rematch support infrastructure;
4. 4D demanded-source channels show measurable relief or controlled redistribution, especially in radial-null, angular-pressure, and current channels;
5. point peaks and angular-pressure leakage remain bounded under grid refinement;
6. shell-throat concentration is stable under matched-strength timing and radial-shape perturbations;
7. release, carve, lapse, packet beta-rematch, and packet-local radial-support derivative diagnostics remain bounded under grid refinement;
8. the source-target decomposition assigns the dominant non-live radial body to a realizable radial-tension scaffold;
9. the endpoint/junction layer is represented by a finite-width source package with bounded reset-cap support, finite support-edge closure, and live exclusion;
10. the regulated endpoint medium has a small current-regulator budget, positive transport margin, and constrained angular response without hidden live, collar, packet, current, or extra closure components;
11. affine-reparameterized radial-null source-sector windows retain positive benchmark margin with non-affinity and geodesic residual monitoring;
12. dense branch-band finite-bundle diagnostics preserve ray ordering, recovery, radial escape, and bundle-width floor through the carrier-focus collar;
13. exterior-null service-time rating gives a positive operational A-to-B advantage under the selected prescribed schedule;
14. chronology-governed service sequencing maps accepted service edges, return edges, reset intervals, and network cross-links into monotone rail-time advance with safety margin;
15. the component can be stated as a continuous metric expression with clear source-role and causal-carrier assignments.

16.3 Acceptance Standard

A metric family is accepted as a load-bearing embodiment when it satisfies all of the following:

1. live packet norm is safely negative under the selected service factor, with edge-load checks providing stress evidence;
2. packet $\Delta\rho$ and packet-comoving negative-density increments remain quiet;
3. incremental ADM momentum is localized in catch/rematch support infrastructure;
4. 4D demanded-source channels show measurable relief or controlled redistribution, especially in radial-null and current channels;
5. point peaks and angular-pressure leakage remain bounded under grid refinement;
6. shell-throat concentration is stable under matched-strength timing and radial-shape perturbations;
7. release, carve, lapse, packet beta-rematch, compact handoff, and packet-local radial-support derivative diagnostics remain bounded under grid refinement;
8. the source-family target grammar closes the major non-live radial support, endpoint/junction, angular-capacity, current-relaxation, and live-handoff sectors;
9. the beta075 endpoint source is represented by a frozen finite endpoint package and a regulated anisotropic heat/current medium passing constrained finite-difference field-closure diagnostics;
10. affine-reparameterized radial-null SNEC diagnostics retain positive benchmark margin under the selected source-sector package;
11. causal-carrier diagnostics show service-stage packet protection and live-service radial escape under the selected prescribed-metric probes;
12. dense finite-bundle carrier diagnostics show branch-band recovery with zero caustic-like bundle-collapse flags under the selected prescribed-metric probes;
13. exterior-null service-time ledgers show operational A-to-B advantage under schedule-factor and packet-coordinate transport proxies;
14. chronology-governed service sequencing assigns operational superluminality to a rail-time monotonicity governor and reserves network-scale synchronization as a future safety-engineering task;
15. reset-domain wake-tail monitors remain exterior to live service and are tracked separately from carrier-band branch behavior;
16. the component can be stated as a continuous metric expression with clear source-role assignments.

17. ENGINEERING ACCEPTANCE CRITERIA

Criterion	Meaning	Gate
Packet norm safety	All live packet norm samples remain negative	Required
Substrate-separated density	Packet $\Delta\rho$ is small after subtraction	Required
Handoff localization	Service Δj_l concentrates in catch/rematch and support-shell handoff	Required
Reduced support target	Selected support-shell target preserves catch/support fraction, rich packet quietness, and partition closure	Required
Continuous metric expression	Support-shell component is represented directly in α, β, G, Q as a continuous metric component	Required
Matched shell strength	Shape comparisons are made at equal peak $\Delta\beta_{\mathrm{sh,max}}$	Required for shape selection

Radial bearing shape	Raised-cosine annular support shell is preferred under matched-strength comparisons	Required for support-shell source comparator
Release choreography	Matched-hold, smooth beta fade, coordinated lapse/carve support, and packet-edge beta rematch preserve packet safety	Required for representative V5 embodiment
Packet beta-rematch sleeve	Smooth blended floor and widened trailing-edge sleeve survive grid refinement with zero high-burden live top points	Required for representative V5 embodiment
Compact handoff layer	Endpoint-smooth entry/catch/edge windows separate pressure containment from derivative smoothing	Required for representative V5 embodiment
Affine radial-null margin	Affine-reparameterized finite-window radial-null exposure is above the selected benchmark floor on live, handoff, endpoint, and source-sector scopes, with non-affinity and radial residual monitors recorded	Required for source-family accounting
4D demanded-source behavior	Radial-null, radial-current, pressure, and packet-comoving density channels remain bounded and improve or redistribute under coupled timing	Required for load-bearing acceptance
Source-family target grammar	The demanded-source ledger decomposes into radial string-cloud backbone, endpoint/junction layer, angular-capacity support, live handoff trim, and current relaxation	Required for source-family realization
Endpoint/junction closure	Endpoint shoulder and reset/decompression cap are represented by a bounded reset-cap body plus finite support-edge closure component with transfer, conservation, and dense-stability diagnostics	Required for endpoint-source acceptance
Endpoint current regulator	A small non-live $\rho + p_l$ regulator removes flux-dominant radial-block rows without adding current, angular, collar, packet, or extra closure components	Required for beta075 endpoint medium
Regulated endpoint medium closure	The regulated anisotropic heat/current medium carries the radial heat/current block and internal angular response with positive transport margin and zero live, regulator-live, or superluminal rows	Required for finite-difference physical-source promotion
Beta-memory receiver	Release-transfer memory is routed into a localized non-live endpoint receiver with packet exclusion preserved	Required for representative receiver grammar
Causal carrier reachability	Entry reachability, radial escape, and scheduled probe diagnostics preserve the live service corridor	Required for horizon/carrier acceptance
Measured cone classification	Local GZ-like branch behavior is measured as a carrier-band feature and classified by reachability, escape, and scheduled-probe gates	Required for carrier acceptance

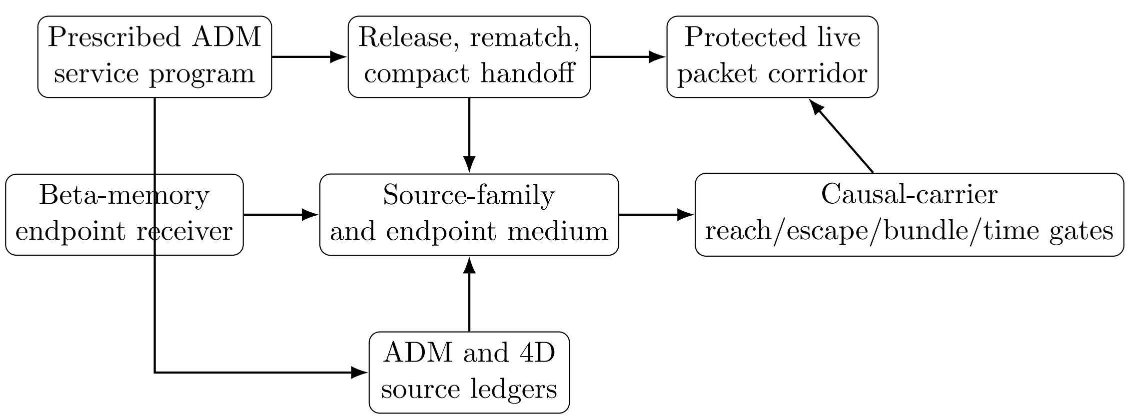
Dense finite-bundle transport	Branch-band null bundles preserve radial escape, recovery from both-shrinking intervals, ray ordering, and bundle-width floor under the beta-collar rematch setting	Required for carrier acceptance
Service-time advantage	Reconstructed restored packet arrival beats the exterior-null reference under schedule-factor and packet-coordinate proxies	Desired for operational rating
Rail-time chronology governor	Service edges, return links, cross-links, reset intervals, and ordinary causal edges advance a common rail time with margin	Required for service-network safety engineering
Wake-tail restoration	Reset-decompression packet-geometry tails remain exterior to live service and are monitored by future-domain closure tests	Required for reset-domain acceptance
Residual localization	Residual maps show localized packet-edge and support-region structure	Required
$V = 5$ stability	Primary embodiment is stable under grid and seed variation	Required
$V = 10$ edge behavior	Edge-load case intensifies the handoff channel and characterizes the same architecture at higher service load	Desired
Compact absorber law	Catch-edge absorber is expressible as a bounded control law	Required
Constraint compatibility	Coarse 3+1 scaffold maintains stable constraint behavior	Required

18. RISK CONTROLS AND DIAGNOSTIC CONTROLS

1. **Packet-region density increment.** Packet-region $\Delta\rho$ concentration is controlled by packet-corridor shaping.
2. **Refinement-sharpening catch layer.** Catch-edge sharpening under grid refinement is controlled by absorber broadening and source-family smoothing.
3. **Angular-pressure leakage.** Growth in p_Ω or angular support entering the packet region is controlled through the throat-capacity/angular jacket sector.
4. **Momentum delocalization.** Δj_l spreading across the packet interior is controlled by the handoff model and entry service gate.
5. **Single-channel additive burden.** Carrying-flow amplitude that adds radial-null or radial-current burden is paired with coupled clock-lapse, rail-stretch, or throat-capacity refinement.
6. **Shell-throat overlap concentration.** Added radial-null, radial-current, or angular-pressure burden where W_{sh} overlaps G, Q gradients is controlled by timing broadening and intrinsic-metric partner refinement.
7. **Point-peak growth.** Local peak growth is controlled by timing broadening, strength selection, and source-family smoothing.
8. **Radial bearing over-broadening.** Support-shell annulus spread into packet-sensitive structure is controlled by the localized raised-cosine bearing band and narrower comparator families.
9. **Packet beta-rematch seam dependence.** Floor/sleeve transition sensitivity is controlled by smooth blended rematch shaping.

10. **Smooth-rematch margin loss.** Narrow trailing-edge support is controlled by radial rematch broadening and pressure-aware handoff smoothing.
11. **Edge-load qualitative transition.** $V = 10$ source-pattern category changes are treated as edge-load classification data for the same architecture.
12. **Radial scaffold realization.** The dominant non-live radial infrastructure burden is controlled by expressing it as a constant-flux radial-tension backbone with explicit residual trims.
13. **Endpoint/junction concentration.** Support-edge shoulder and reset/decompression cap burdens are controlled by a frozen finite endpoint source package with bounded reset-cap support, finite support-edge closure, transfer accounting, concentration checks, and dense-stability diagnostics.
14. **Endpoint current and angular response.** Flux-dominant radial-block rows are controlled by a small non-live current regulator, while the non-live angular-pressure watch is controlled as an internal response of the same regulated anisotropic heat/current medium.
15. **Receiver timing.** Beta-memory endpoint receiver action is controlled by release-transfer memory, negative-l localization, packet exclusion, and service-factor retiming anchors.
16. **Causal carrier branch structure.** Local radial branch behavior is controlled by reachability, radial escape, shell/throat overlap, scheduled-ADM probe diagnostics, and dense finite-bundle carrier audits.
17. **Carrier-focus collar compression.** Finite-bundle compression in the minus-branch carrier band is controlled by widening and timing the trailing-edge beta-rematch collar until dense bundle transport remains above the caustic-like collapse thresholds.
18. **Operational service-time rating.** A-to-B service timing is controlled by an exterior-null comparison ledger that records prepared-service and request-triggered service clocks, schedule-factor distance, packet-coordinate distance, and the strict $l = \sigma$ centerline control.
19. **Chronology-governed service sequencing.** Service arming is controlled by a rail-time governor that advances entry, carrier, return, cross-link, reset, endpoint-readiness, and multi-leg routing intervals with configured margin.
20. **Wake-tail restoration.** Reset-domain packet-geometry tails are controlled by explicit post-service wake-tail monitors and domain-extension closure tests.
21. **Constraint stability.** Coarse 3+1 scaffold constraint residuals are controlled through smoothing, reparameterization, and source-ledger redesign.

19. ILLUSTRATIVE SYSTEM DIAGRAM



20. CLAIM-ORIENTED EMBODIMENTS

The following claim-oriented embodiments clarify the technical scope of the disclosure.

1. A method of operating a prescribed spacetime metric service, comprising preparing a standing support substrate, defining a live packet corridor within a larger support shell, routing a carrying-flow field through the support shell, and applying a locked catch/rematch control window to localize service-induced momentum demand near a packet edge.
2. The method of item 1, wherein the live packet corridor has radius R_{pass} smaller than a support radius R_{th}, and wherein a radial support-edge width w_{th} is selected to reduce radial pressure burden while a lapse cushion preserves packet causal margin.
3. The method of item 1, wherein a dimensionless service factor V sets the scheduled carrying-flow level and packet-frame comparison schedule, and wherein V is used to compare primary and edge-load embodiments.
4. The method of item 3, wherein a primary operating embodiment is rated at $V = 5$ and an edge-load operating embodiment is rated at $V = 10$.
5. The method of item 1, wherein an ADM diagnostic ledger is computed from fields α, β, $G = \gamma_{ll}$, and $Q = \gamma_{\Omega\Omega}$, and wherein the ledger includes ρ_{ADM} and j_l.
6. The method of item 5, further comprising subtracting a time-matched standing substrate from a live service field to identify $\Delta\rho$ and Δj_l.
7. The method of item 6, wherein a service pass is accepted when $\Delta\rho$ is small inside the live packet and Δj_l is localized near a catch/rematch handoff layer.
8. A spacetime metric engineering system comprising a standing support shell, an angular jacket, a clock-lapse cushion, a support-contained carrying-flow field, a live packet corridor, and a packet-edge catch/rematch absorber.
9. The system of item 8, wherein the packet-edge catch/rematch absorber comprises a kinematic handoff law multiplied by a compact inner-edge spatial envelope centered near $\xi = -R_{\text{pass}}$.
10. The system of item 9, wherein a residual-responsive limiter bounds the absorber amplitude.
11. The system of item 8, wherein a support-shell overlay is applied through an annular timing window W_{sh} and modifies one or more of β, α, G, and Q.
12. The system of item 11, wherein the support-shell overlay includes

$$\beta = \beta_0 + a_\beta W_{\text{sh}}, \quad \alpha = \alpha_0 e^{\kappa_\alpha W_{\text{sh}}}, \quad G = G_0 e^{\kappa_G W_{\text{sh}}}, \quad Q = Q_0 e^{\kappa_Q W_{\text{sh}}}.$$

13. The system of item 12, wherein W_{sh} includes a raised-cosine annular radial bearing function centered in the support-shell infrastructure band.
14. The system of item 13, wherein support-shell shape comparisons are normalized by a matched peak carrying-flow perturbation $\Delta\beta_{\text{sh,max}}$.
15. The method of item 1, further comprising a release-choreography law that holds the packet in a matched state, fades beta with a finite smooth profile, maintains lapse/carve support during the fade, and relaxes support after packet-edge mismatch is reduced.
16. The method of item 15, wherein a packet-local beta rematch correction is applied through a trailing-edge sleeve and a smooth blended center floor.
17. The method of item 16, wherein the trailing-edge sleeve is radially widened until the blended floor survives grid refinement with all live packet-norm samples safely negative and zero top-channel live packet points at the primary service factor.
18. The method of item 15, further comprising a compact handoff layer with endpoint-smooth entry containment, catch/rematch bearing, and trailing-edge smoothing windows.
19. The method of item 18, wherein the compact handoff layer assigns live radial-pressure containment to the entry window and angular/current plus radial-null derivative relief to catch and edge windows.
20. The system of item 8, further comprising a packet-local radial support profile with a mild filled core stretch, a narrow annular catch ring, and a weak annular skirt in γ_{ll}.
21. The system of item 8, further comprising a source-family target grammar that assigns demanded-source burden to a constant-flux radial string-cloud backbone, a coupled endpoint/junction layer,

small core-body residual leakage, live handoff trim, angular-capacity support, and non-live current relaxation.

22. The system of item 21, wherein the constant-flux radial string-cloud backbone has positive energy density, radial tension close to $p_l = -\rho$, small transverse pressure, small current, and a nearly constant areal flux proxy $\Phi/\gamma_{\Omega\Omega}$.
23. The system of item 21, wherein the endpoint/junction layer carries support-edge shoulder trim and reset/decompression cap demand in one coupled finite-width source target.
24. The system of item 23, further comprising a beta-memory endpoint receiver whose amplitude is tied to accumulated beta-release transfer and whose active support is localized in a non-live negative-l support-edge receiver band.
25. The system of item 24, wherein a beta100 edge-load receiver setting uses the same receiver grammar with service-specific post-release retiming to preserve selected, current, and angular endpoint transfer accounting.
26. The method of item 1, further comprising affine-reparameterized radial-null accounting using a coordinate-time radial null tangent, measured non-affinity, affine window weighting, and radial geodesic residual monitors.
27. The method of item 1, further comprising causal-carrier accounting using radial null speeds $v_\pm = -\beta \pm \alpha/\sqrt{\gamma_{ll}}$, branch-margin maps, measured cone classification, entry reachability, radial escape, and scheduled probe evolution.
28. The method of item 27, wherein reset-domain packet-geometry wake-tail traces are monitored separately from live-service carrier acceptance and remain exterior to the live packet corridor during the scheduled probe audit.
29. The method of item 27, further comprising a dense finite-bundle carrier audit in which recovered branch-band seeds are expanded into radial null bundles, and wherein a trailing-edge beta-rematch collar is widened or timed until the bundles preserve radial escape, recovery from both-shrinking intervals, ray ordering, and a positive bundle-width floor.
30. The method of item 29, wherein the dense finite-bundle carrier audit is treated as a CHL-adjacent branch-band optics gate for the throat-loaded active-rail service architecture.
31. The method of item 27, further comprising an exterior-null service-time advantage ledger in which exterior endpoints A and B are fixed, packet departure and restored arrival events are recorded, prepared-service and request-triggered clocks are evaluated, and schedule-factor and packet-coordinate transport proxies are compared with an exterior null reference time.
32. The method of item 31, further comprising a rail-time chronology governor in which each armed service edge, return edge, cross-link, reset interval, and ordinary causal edge advances a common rail time with safety margin, and in which connected rail networks are scheduled by one chronology policy.
33. The system of item 23, wherein the endpoint/junction layer is represented by a frozen finite endpoint source package comprising a bounded reset-decompression cap body and a finite support-edge closure component, each excluded from the live packet corridor.
34. The system of item 23, further comprising a non-live endpoint current regulator applied in the radial $\rho + p_l$ pair, wherein the regulator makes a radial heat/current block boost-diagonalizable without adding an independent current, angular, collar, packet, or conservation-closure component.
35. The system of item 23, wherein the endpoint/junction layer is carried by a regulated anisotropic heat/current medium with an internal angular-pressure response and a positive transport-margin diagnostic under finite-difference field-closure validation.
36. A numerical source-accounting process comprising constructing an exact field builder for α, β, G, and Q; computing ADM constraint fields; performing standing substrate subtraction; computing a 4D demanded-source ledger; recording release/carve/lapse/beta-rematch/compact-handoff derivative diagnostics; computing semilocal finite-window null-accumulation margins; extracting source-

family targets; evaluating endpoint/junction receiver transfer; classifying endpoint source blocks; applying a non-live current regulator; validating a regulated endpoint heat/current medium; and evaluating source-family realizations against component-specific source roles.

21. DIAGNOSTIC DATA PRODUCTS

Data products for the primary $V = 5$ operating embodiment include:

1. exact-builder ADM implementation package;
2. substrate-subtracted ρ_{ADM} and j_l maps;
3. reduced support-shell ansatz ledger and selected support-shell target ledger;
4. generalized diagnostic ladder with target/ramp separation and service-factor input generation;
5. V10 selected-target edge-load comparison;
6. continuous support-shell 4D demanded-source ledger path;
7. coupled timing sweep for carrying-flow, clock-lapse, and rail-stretch support-shell partners;
8. throat-capacity same-window ledger;
9. support-shell shape-selection and matched-strength robustness ledger;
10. release-choreography ledger;
11. release-choreography floor-union refinement check showing preservation of the blended, widened trailing-edge V5 family under higher grid resolution;
12. compact handoff, entry service gate, and channel-cause ledgers for pressure-aware entry/catch/edge shaping;
13. full-grid source-decomposition and dense source-sector closure ledgers;
14. source-family target extraction for the constant-flux radial string-cloud backbone, endpoint/junction layer, residual leakage, angular capacity, live handoff trim, and current relaxation;
15. beta-memory endpoint receiver transfer-accounting and retiming ledgers;
16. frozen endpoint-J source ledgers for the bounded reset-cap body and finite support-edge closure component;
17. endpoint source-class, current-regulator, regulated-medium admissibility, and constrained field-closure ledgers;
18. causal-carrier reachability, radial escape, shell/throat branch-margin, scheduled-ADM probe, and dense finite-bundle carrier ledgers;
19. tensor-consistent beta-collar generator ledgers for the conservative $rematch\_w6\_t1p5$ carrier lead and the wider $rematch\_w8\_t2p0$ optical-relief bracket;
20. exterior-null service-time advantage ledgers comparing prepared-service and request-triggered service clocks with schedule-factor and packet-coordinate transport proxies;
21. chronology-governance source materials and future service-sequencing safety specifications for rail-time monotonicity across single-rail and connected-network operation;
22. affine-reparameterized radial-null SNEC and semilocal finite-window null-accumulation diagnostics comparing selected radial-null exposure windows with a selected benchmark floor and residual monitors.

22. CONCLUSION

The active-rail system organizes spacetime metric engineering as a source-placement and causal-carrier architecture with an explicit source-family target grammar. The primary $V = 5$ operating embodiment separates standing substrate curvature from live service demand and localizes selected support-shell and handoff increments into catch/rematch, support-edge, and endpoint infrastructure while preserving packet safety. The $V = 10$ operating embodiment supplies an edge-condition comparison for the same architecture, while beta100 receiver retiming supplies a release-timing stress anchor.

The refined metric understanding is componentized: carrying-flow β expresses scheduled service load, clock-lapse α protects timing and causal margin, rail-stretch G controls radial metric capacity, and throat-capacity Q carries angular/throat support. The continuous support-shell overlay is written as a metric-side family. Demanded-source results retain raised-cosine annular support-shell bearing and clock-lapse coupling as important source-ledger comparators, while the representative end-to-end V5 operating embodiment is the matched-hold release choreography with a smooth blended trailing-edge beta-rematch sleeve, finite beta-collar widening for branch-band bundle transport, two-feature packet-local radial support profile, entry service gate, compact pressure-aware handoff layer, source-sector closure, beta075 $p003\_mid$ receiver accounting, frozen endpoint source-medium closure, affine-reparameterized radial-null ledger, exterior-null service-time rating, and rail-time chronology governance.

The algebraic ledger shows a structured source-placement picture. The highest live angular/current rows are packet-in-support rows where packet carve and beta rematch are active ahead of the radial core/ring/skirt windows. The entry service gate places setup support in infrastructure accounting and starts live packet scoring inside the supported corridor. The compact handoff layer improves broad semilocal radial-null and angular-pressure exposure, while pressure-aware entry shaping supplies a direct channel for live radial-pressure control. The non-live radial body is represented by a constant-flux radial string-cloud backbone, and the remaining support-edge plus reset/decompression demand is carried by a coupled endpoint/junction layer with beta-memory receiver accounting, a bounded reset-cap body, a finite support-edge closure component, and a regulated anisotropic heat/current medium. The causal-carrier diagnostics place the measured GZ-like cone signature, CHL-adjacent branch-band optics, entry reachability, branch-band radial escape, dense finite-bundle carrier transport, exterior-null service-time advantage, rail-time chronology governance, scheduled ADM probe evolution, and wake-tail restoration in separate acceptance gates. Thus the V5 architecture has robust packet safety, localized demanded-source structure, explicit source-family realization targets, a first finite-difference beta075 endpoint physical-source candidate, positive finite-window radial-null margins, favorable operational service-time rating, rail-governed chronology policy, and a prescribed-metric carrier interpretation for the branch-sensitive shell/throat band.

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